

Single-cell foundation models represent genes in context, but reveal no context-specific gene function beyond co-expression

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Abstract

Background: Single-cell foundation models build, at every transformer layer, a representation of each gene that depends on the other genes in the cell. Whether this contextual representation shows how a gene's role differs between cell types, the single-cell counterpart of word sense in language models, has not been tested.

Results: We measured contextual gene representations in MaxToki (217M and 1B), scGPT and STATE on identical Tabula Sapiens cells from 12 cell types, and in MaxToki-217M also on independent bone-marrow data. In every model a gene's representation changes with cell type in a gene-specific way: the change agrees between two independent halves of the cells, with a cosine 0.74 to 0.87 higher than for a randomly paired gene. It is not explained by the gene's

rank in the cell and replicates on the bone-marrow data. Untrained versions keep 36–88% of it, and a representation built from expression alone also shows it (+0.321). In MaxToki-217M the change follows functional axes such as nuclear versus surface genes ($z = +13.3$ against random axes; on this axis STATE shows less and scGPT little). In all four models it follows co-expressed gene sets. None of three expression-only representations shows either kind of organisation. In no model is the functional organisation significantly stronger than that of gene sets matched on co-expression (smallest $p = 0.07$), although this test detects only large differences. Adding the nuclear/surface direction to half of a cell's genes shifts MaxToki's predictions for the other half more than random directions from the same genes; as the pushes exceed natural changes, this shows the model can read the direction, not that it uses it. In both MaxToki models, genes that change more gain slightly more from their own context in next-gene prediction. Two direct tests for context-specific gene function were negative in all four models.

Conclusions: Single-cell foundation models build real contextual gene representations, organised by co-expression. We found no context-specific gene function beyond co-expression, though our tests could detect only clear effects. Models trained on more than expression were not tested. Functional organisation separates some architectures but does not rank them reliably.

Keywords: Single-cell foundation models; Contextual gene representations; Mechanistic interpretability; Gene co-expression; Gene embeddings; Transformers; Single-cell RNA sequencing; Model evaluation

Background

A recurring theme in the interpretability of large language models is that a network trained on a narrow predictive task builds internal representations of structure it was never taught. Sequence models trained on Othello and chess build internal representations of the board and of hidden variables such as player skill [1–3]. Language models encode space and time along single directions that a linear probe can read [4]. In the best cases the model also uses these representations to make its predictions. One tests this by changing the representation and measuring the effect on the model’s output [5–7].

A closely related line concerns contextual representations. Contextual word embeddings assign the same word a different vector in every sentence, and the geometry of those vectors encodes word sense: “bank” near a river and “bank” in finance occupy separable regions [8–10]. These measurements need care, because representation spaces are strongly anisotropic (vectors cluster in a narrow cone rather than spreading evenly, so any two look similar) and a handful of “rogue” dimensions can dominate cosine similarity [9, 11–13].

Single-cell foundation models (SCFMs) offer a clean biological version of the same question [14–17]. They treat each gene as a token and each cell as a set or sequence of genes, either ordered by expression (Geneformer) or paired with binned or continuous expression values (scGPT, scBERT, scFoundation), and are trained to predict masked or next genes or values, in effect learning which genes tend to be active together. Their gene token embeddings (one fixed vector per gene) have been studied [15, 18]; their contextual gene representations (a gene’s vector inside a particular cell) are used as features downstream [16, 19]. Their geometry has been mapped when averaged over cells or within one tissue (Related work), but how a single gene’s vector changes from one cell type to another has not been measured. A model could use this

vector to represent that a gene plays a different role in a macrophage than in a T cell. That would be the biological counterpart of word sense. The “cell as a token” framing points to this question but does not test it [20].

Related work. Our methods descend directly from the language-model literature on contextualised embeddings. Ethayarajh [9] showed that contextual representations are anisotropic in all non-input layers, that same-word self-similarity falls in upper layers, and that every cosine must be corrected against an anisotropy baseline. Timkey and van Schijndel [13] showed that a few rogue dimensions dominate similarity and that a scalar correction is not enough; Mu et al. [11] and Cai et al. [12] give the standard postprocessing and the local-isotropy picture. Coenen et al. [10] and Garí Soler and Apidianaki [21] show that word senses form separable clusters and that a word’s degree of polysemy is itself readable. We import both the measurements and the controls. From the literature on latent variables that models use causally [1–4] we adopt one rule: a probe shows only correlation, so a causal claim needs an intervention compared against a matched control. We also use the steering methods of this literature [5, 6]. Templeton et al. did the same at large scale, extracting interpretable feature directions from a production model and steering it with them [22].

Geneformer [14], scGPT [15], scFoundation [16], scBERT [17], Universal Cell Embeddings (UCE) [23] and STATE [24] differ in tokenisation (rank-ordered gene identity, binned or continuous expression values, or gene tokens derived from protein-sequence embeddings), and the review of Bian et al. [19] defines the “contextual gene embedding” that our work dissects. Max-Toki [25], the primary model here, is a Llama-architecture, Geneformer-lineage model at 217M and 1B scales. Two earlier analyses come closest to a “context changes the gene” claim. scGPT reports cell-state-specific gene networks read from attention [15], but as a pairwise network, not as a property of the representation vector, and without a decomposition or a null. Geneformer’s

in silico deletion measures how the embeddings of the other genes shift when one gene is removed from a cell [14], which is a change of context by one gene. Neither characterises how a gene’s representation differs between natural cell types. The geometry of gene representations in these models has been mapped, both for static gene embeddings [18] and for layer-wise hidden states averaged over cells, including their spectral structure [26] and the topological and geometric structure they do and do not encode, per tissue [27]. Residual-stream gene vectors built separately in several tissues carry a small, tissue-dependent amount of transcription-factor-target signal beyond expression covariates [28]. That test used TRRUST edges, which are the same in every tissue, so it did not ask whether a gene’s role differs between tissues.

A critical literature finds that SCFM’s often fail to beat simple baselines: on zero-shot tasks [29], under deeper evaluation [30], on perturbation prediction [31, 32], and across utilities [33]. Sparse autoencoders find organised biological knowledge but minimal regulatory logic in these models [34], and edge scores read from SCFM attention add nothing beyond simple gene-level statistics for predicting which genes respond to a perturbation [35]. Co-expression is the canonical guilt-by-association signal [36], and some gene-embedding methods, such as Gene2vec, are built directly on it [37]; others build gene embeddings from text descriptions of genes instead [38]. Genuine context-dependent gene function exists. Some of it acts at the protein level, as when a moonlighting protein changes function with its location or binding partners [39], and need not change the gene’s expression. Some acts through transcription, as when lineage-determining transcription factors select different enhancers with different partners by cell type [40].

Aim. We organise the question into two levels. Level 1 (a fact about the model): is a gene’s representation reshaped by context in a structured, gene-specific way, is that reshaping learned, and is it linked to what the model does? Level 2 (biology): does the reshaping reveal something

111 about the gene’s role in each context beyond what co-expression already tells us? We study
112 Level 1 for MaxToki-217M in depth (rank control, independent replication, random-weights
113 control, link to the model’s predictions, steering) and repeat the central tests in MaxToki-1B,
114 scGPT and STATE on identical cells. For Level 2, we test the functional organisation of the
115 contextualisation against co-expression nulls and against representations built from expression
116 alone, and run two direct tests for context-specific gene function, in all four models. The study
117 makes three contributions. (1) It measures how much, and how gene-specifically, a gene’s rep-
118 resentation changes between cell types (Level 1). It also tests whether the model reacts when
119 functional directions are pushed into it, and whether genes that change more gain more from
120 their own context in next-gene prediction. Neither test shows that the natural change is used.
121 (2) It shows that this change is organised by co-expression, and that two tests find no context-
122 specific gene function beyond co-expression in any of the four models (Level 2). (3) It tests
123 whether the functional organisation of the change can be used to compare models, and where
124 that comparison breaks down.

125 **Methods**

126 **Models**

127 MaxToki-217M and MaxToki-1B [25, 41] are Llama-architecture autoregressive models over
128 rank-ordered gene tokens, with 11 and 20 transformer layers and hidden widths of 1232 and
129 2304. We used the Hugging Face safetensors checkpoints through the transformers library
130 with `output_hidden_states=True`, in float32. For MaxToki, “layer k ” is the k -th hidden state:
131 layer 0 is the embedding output and layer $k \geq 1$ is the output of the k -th transformer block; the last
132 hidden state (layer 11 of MaxToki-217M) is taken after the model’s final normalisation. scGPT

[15, 42] (whole-human checkpoint; 12 layers; width 512) and STATE [24, 43] (the SE-600M state-embedding model; 16 layers; width 2048) were used for the cross-architecture comparison. STATE’s gene tokens are fixed protein embeddings from the Evolutionary Scale Modeling 2 (ESM2) language model [44], passed through a learned projection together with each gene’s expression. For all models, “layer k ” is the representation after k transformer blocks, so models can be compared at matched relative depth (k divided by the number of blocks). As untrained controls, we built each architecture with random initial weights (the library’s default initialization, seed 0): a LlamaForCausalLM from MaxToki-217M’s configuration with the same vocabulary, scGPT’s TransformerModel with the whole-human configuration, and STATE’s model with all weights random except its gene table. STATE never trains this table (fixed ESM2 protein embeddings), so it is the same in the trained and untrained models; the learned projection that reads it is random.

Datasets and annotations

The main data are the Tabula Sapiens [45] immune, kidney and lung datasets from CZ CELLx-GENE Discover [46–49]. For the immune dataset we used a 20,000-cell subset; its cell identifiers are listed in the code repository. These files hold $\log(1 + \text{counts per } 10,000)$, written $\log_{10}$ below, in their main matrix, computed from counts corrected for ambient RNA, and the original integer counts in their raw matrix. All model inputs, the expression-only representations and the co-expression matrix were computed from the raw counts, so they include ambient RNA. Most selected cells come from 10x 3’ assays; about 4% are Smart-seq2 cells, whose counts are read counts. Contexts are cell types, with cells pooled over the three datasets: the 12 cell types with the most cells (macrophage, pulmonary alveolar type 2 cell, kidney epithelial cell, capillary endothelial cell, CD4-positive alpha-beta T cell, CD8-positive alpha-beta T cell, B cell, basal

cell, pulmonary alveolar type 1 cell, intermediate monocyte, neutrophil and classical monocyte), each with at least 1000 cells. We took 1000 randomly chosen cells per cell type (seed 0). Some cell types come from one dataset and others mix two or three (for example, most macrophages are lung cells), so a shift between two cell types can also reflect a different tissue mix.

For independent replication we used the human CD34+ bone-marrow data of Setty et al. [50] (5780 cells) as distributed with scVelo [51, 52], which holds velocity spliced and unspliced integer counts for 14,319 pre-filtered genes (11,842 in the MaxToki vocabulary); its main matrix holds the spliced counts only. Because most Tabula Sapiens counts come from 10x assays that also count intronic reads, the main Setty analysis ranks genes by the sum of spliced and unspliced counts; a robustness run used spliced counts only. Contexts are the nine differentiation clusters with at least 130 cells: two haematopoietic stem cell clusters, precursors, common lymphoid progenitors, two erythroid clusters, two monocyte clusters and dendritic cells. Megakaryocytes (61 cells) were excluded, and all 5719 cells of the nine kept clusters were used.

Gene Ontology (GO) annotations [53, 54] were taken from the gene-to-GO file distributed with GEARS [55, 56]. A gene belongs to a functional pole only if it is annotated directly to one of the listed GO terms; annotations were not propagated through the GO hierarchy. Curated transcription-factor (TF) targets were all human TRRUST v2 interactions [57, 58] together with DoRothEA interactions of confidence A or B [59, 60]. Gene symbols were mapped to Ensembl identifiers with the Geneformer dictionaries (version gc104M) [14, 61].

Input encoding and extraction of contextual gene representations

MaxToki uses Geneformer’s rank-value encoding [14]: a cell’s counts are normalised to the cell total, each gene is divided by its median from Geneformer’s gc104M gene-median dictionary (MaxToki’s vocabulary holds the same 20,271 genes; the released MaxToki checkpoints include

no median dictionary of their own), and the non-zero genes are ranked in decreasing order. A per-cell scale factor does not change the order within a cell, so the order is that of raw count divided by gene median. The first 1022 genes were kept and wrapped in the beginning- and end-of-sequence tokens (1024 tokens at most), a quarter of the model’s 4096-token limit, to limit compute. Most cells express more genes than this, so MaxToki saw roughly the top 40% of a typical cell’s genes, ranked by count divided by gene median. On real cells, this ordering gives the same token sequences as a separate rank-value tokenizer we wrote earlier for MaxToki, except that genes whose count-to-median values are equal after rounding to 32-bit floating point can appear in swapped order; we did not compare against an official MaxToki tokenizer.

We recorded, for each gene in a panel, its hidden state at that gene’s token position and averaged it over the cells of one cell type. The main MaxToki-217M extraction used layers 0, 4, 8 and 11. A separate extraction of all 12 hidden states (the embedding output and all 11 transformer layers) with a 4000-gene panel, on the same cells, gave the full depth profile. Four design choices guard against known sources of false signal. (1) Pairs of cell types, not only genes found in every cell type: each per-gene mean is built from at most 50 occurrences in each half of the cells (see (3)), and we call a gene count-balanced in a cell type if it reaches that cap in both halves. In the main extraction only 137 genes are count-balanced in all 12 cell types, many of them broadly expressed housekeeping genes (the single-cell analogue of the “stopwords” that are most context-sensitive in language models [9]), whereas each pair of cell types shares a median of 1391 count-balanced genes. So we compare cell types two at a time. The panel contains genes with at least 25 occurrences in at least two cell types (the 6000 most frequent such genes). (2) Cell-level split halves: the cells of each cell type were split into two equal halves (500 cells each in the main extraction, 300 each in the 600-cell runs). So every per-gene mean is estimated twice, from different cells, and a result counts only if it repeats in both halves. (3) An occurrence cap:

203 a per-gene mean has a standard error proportional to $1/\sqrt{n}$, and unequal counts inject context-
204 dependent noise into any interaction term, so at most 50 occurrences were accumulated per
205 (gene, cell type, partition), over a shuffled cell order. A (gene, cell type) entry is used only if it is
206 count-balanced, that is, it reached the cap in both partitions. (4) Per-dimension z-scoring: before
207 EXCESS, the functional axes and every other cosine-based measure, every hidden dimension
208 was standardised over all count-balanced entries, to remove the inflation by a few dominant
209 (“rogue”) dimensions that a scalar anisotropy correction misses [13]. The anisotropy check
210 (below) also reports cosines on the raw representations, for comparison.

211 Other extractions used the same procedure with these settings. Scaling pair: MaxToki-217M
212 and MaxToki-1B on the same 600 cells per cell type with a 5000-gene panel (cap 50), at layers 2
213 and 4 of MaxToki-217M and layers 4 and 7 of MaxToki-1B, so that the models can be compared
214 at matched relative depth (layer 4 of 11 = 0.36 versus layer 7 of 20 = 0.35; layer 2 of 11 = 0.18
215 versus layer 4 of 20 = 0.20). Random-weights control: the same cells and the same 5000-gene
216 panel as the MaxToki-217M scaling run, at layer 4. Setty data: cap 25, a floor of 15 occurrences
217 in at least two clusters, a 5000-gene panel, layers 2 and 4.

218 scGPT and STATE were run on exactly the same cells, cell types and partitions as the 600-cell
219 MaxToki runs, with the same 5000-gene panel and cap 50, each with its own standard input.
220 scGPT (following its checkpoint configuration and its cell-embedding pipeline): the raw counts
221 of all expressed genes in scGPT’s vocabulary were quantile-binned into 50 bins within the cell,
222 1199 of them were then sampled at random if the cell had more, and the <cls> token was
223 prepended; layers 2, 4 and 9 (relative depth 0.17, 0.33 and 0.75). STATE (through its own data
224 collator): raw counts over the genes in its protein table, the top 2047 expressed genes by count;
225 we kept only positions holding expressed genes; layers 3, 6 and 12 (relative depth 0.19, 0.38 and
226 0.75). Of the 5000 panel genes, 4964 are in scGPT’s vocabulary and 4969 in STATE’s protein

table. Each model's own count-balanced entries were used. The untrained controls used the same cells and panel.

Expression-only representations

To test directly whether expression alone produces the functional organisation, we built contextual gene representations with no model, from the same cells, cell types and partitions as the 600-cell runs. Landmark version: in each cell type and partition, gene g was represented by the vector of Pearson correlations, across those 300 cells, between g 's $\log_{10}(\text{counts per } 10,000)$ and that of each of 512 landmark genes (the most variable panel genes over all selected cells). Principal-component version: the same, with the scores of 256 principal components of the z-scored expression of all selected cells in place of the landmarks. Both are correlation profiles within one cell type, so they cannot see genes that rise and fall together between cell types. Our co-expression measure (next sections) pools all cell types and so includes that part. We therefore added a third version, built on axes from all cell types pooled, that we expected to respond to that part. Each of its vectors is still an average over cells of one cell type, divided by the gene's total expression there, so the gene's level in each cell type cancels out; it may therefore not capture co-expression between cell types either, and we did not check this directly. Centroid version: gene g in a cell type and partition was placed at the expression-weighted mean position of its expressing cells in the first 50 principal components of the same pooled, z-scored expression, that is, where in the pooled cell space the gene is expressed within that cell type. As in the model extractions, the cells of each cell type and partition were put in a random order and only the first 50 cells expressing the gene were used, so each vector averages as many cells as a model vector averages occurrences. Without this cap a vector averages all expressing cells (a median of 202 per entry; 10th to 90th percentile 108 to 277); we report that version only

as a check. We first built the centroid version with 512 components and no cap. It failed the positive control: the shift shared by all genes did not replicate between the two halves of the cells (cosine -0.22), because z-scoring each dimension gives the many noisy high-numbered components equal weight. With 50 components, a common default in single-cell analysis, it passes (replication $+0.79$); we did not run other numbers of components through the pipeline. The 512-component and all-cells results are in Additional file 1. The landmark and principal-component versions are correlations across all 300 cells of each half, so they are not matched to the models' 50 occurrences. All three used the same gene panel and the same count-balanced entries as MaxToki-217M and went through the identical pipeline (EXCESS, functional axes, all nulls). "Layer" does not apply to them.

Two-way decomposition and contextualisation strength (EXCESS)

For gene g in cell type c , we write $v(g, c) = \mu + a(g) + b(c) + E(g, c)$: a gene main effect $a(g)$ (the gene's identity), a context main effect $b(c)$ (a shift shared by all genes in that cell type, expected because attention pools over the cell), and a gene-specific interaction $E(g, c)$, the only term where a gene-specific response to context can appear.

For a cell-type pair (c_0, c_1) , the relative shift of gene g is $\delta(g) = [v(g, c_1) - v(g, c_0)] - \text{mean}_{g'}[v(g', c_1) - v(g', c_0)]$: the gene's own shift minus the average shift of all genes. We computed δ separately in the two cell partitions (δ_0, δ_1) and defined $\text{EXCESS} = \text{mean} \cos(\delta_0(g), \delta_1(g)) - \text{mean} \cos(\delta_0(g), \delta_1(\text{perm } g))$: the same gene's shift agreeing across independent cells, minus a gene-shuffled control. $\text{EXCESS} = 0$ means no gene-specific response to context. The means are pooled over all (gene, pair) entries from the cell-type pairs with at least 200 count-balanced genes (150 in the cross-model comparison of contextualisation strength and in the Setty analysis), which equals a mean over pairs weighted by their number of genes. In the

600-cell data three of the 66 pairs have 150 to 199 such genes, so EXCESS in the cross-model comparison can differ by up to 0.003 from EXCESS in the functional-organisation controls. 95% confidence intervals (CIs) come from 2000 bootstrap resamples of genes, in which each resampled gene brings all of its pair entries. As a positive control we report the replication of the context main effect across partitions (the cosine between the two partitions' mean shifts).

Anisotropy. On the raw (unstandardised) representations we computed the mean cosine of random different-gene pairs (the anisotropy baseline [9]), same-gene self-similarity across cell types, and the share of variance on the first principal component (PC), and repeated these after per-dimension z-scoring. Self-similarity was averaged over all genes count-balanced in at least two cell types (5433 at layer 4). The anisotropy baseline is the mean over 6000 random pairs of (gene, cell type) entries, and the first-PC share comes from 4000 randomly drawn entries.

Rank-position control. MaxToki encodes token position, and a gene's rank changes between cell types. We computed each gene's mean rank in each cell type over the cells in which it was among the 1022 genes given to MaxToki (only these cells give the gene a representation), using the same cells as the extraction. We split genes at the median absolute rank change into rank-stable and rank-moving halves and recomputed EXCESS in each, correlated the size of the shift with the rank change, and recomputed EXCESS after a linear regression of the shift on the gene's mean rank in each of the two cell types and on its absolute rank change. Because rank could act in a curved way, we repeated the regression with squared, product and log terms of the ranks. As a second check, we replaced the gene-shuffled control with a rank-matched one: each gene's shift was compared with that of the gene nearest to it in (rank in one cell type, rank in the other). If rank drove the shift, genes with the same ranks would share it.

Functional axes and reproducible interaction power

A functional axis u is the unit vector separating two GO-defined gene classes in the main effect, $u \propto \text{mean}(a(g) \mid \text{class A}) - \text{mean}(a(g) \mid \text{class B})$, where $a(g)$ is the gene's z-scored representation averaged over partitions and count-balanced cell types. Genes annotated to both poles were removed from both. The axes were nuclear versus surface/secreted (nucleus GO:0005634, chromatin GO:0000785 and DNA binding GO:0003677, versus plasma membrane GO:0005886, extracellular region GO:0005576 and extracellular space GO:0005615); mitochondrion (GO:0005739) versus cytoskeleton (GO:0005856); transcription (regulation of DNA-templated transcription GO:0006355 and DNA-binding transcription factor activity GO:0003700) versus transport and signalling (ion transport GO:0006811 and signalling receptor activity GO:0038023); and cell cycle (GO:0007049) versus cell differentiation (GO:0030154). Axis validity is the five-fold cross-validated area under the receiver operating characteristic (ROC) curve (AUC) with which $\langle a(g), u \rangle$ separates held-out pole genes.

Reproducible interaction power measures how much context moves genes along u in a way that replicates across independent cells. For genes count-balanced in at least 9 cell types, we projected each partition's representation onto u , $q_p(g, c) = \langle v_p(g, c), u \rangle$; regressed out the gene's mean rank in that cell type (linear and quadratic terms; "rank-controlled"); took the two-way interaction $I_p(g, c)$ (the projection minus its gene mean and context mean, plus the grand mean); and averaged $I_0(g, c) \cdot I_1(g, c)$ over all (gene, cell type) entries. Noise averages to zero in this product, so only modulation that reproduces across the two partitions survives. Functional-z is the z-score of this power against 300 axes built from random gene partitions of the same pole sizes. We report it at layers 4, 8 and 11, without and with the rank regression. A cell type's loading on an axis is the mean projection onto u of the z-scored layer-4 representations (averaged over partitions) of the genes count-balanced in that cell type. These gene sets differ

between cell types and the loadings are not rank-controlled, so we use them only to describe the ordering of cell types.

Null models for functional organisation

Functional gene sets may be co-expressed, and co-expressed genes co-vary across cells whatever the model “understands”. We therefore compared each of the three axes with well-separated poles (nuclear/surface, mitochondrion/cytoskeleton and transcription/transport; the cell-cycle/differentiation axis separates its poles only weakly and was not tested) with null axes whose two poles have the same sizes as the functional poles, using the rank-controlled power. Null 1 (random axes) is the null behind functional-z and was computed at layers 4, 8 and 11; its empirical p was computed at layer 4. Nulls 2 and 3 were computed at layer 4, and null 4 at layers 4 and 8. In the 600-cell comparison each model was tested at its own depth-matched layer. Co-expression was the Pearson correlation of $\log_{10}(\text{counts per 10,000})$, computed from the raw counts over 3000 cells from each of the three Tabula Sapiens datasets, and the coherence of a gene set was its mean pairwise correlation.

1. **Random axes:** random genes with the functional poles’ sizes (300 axes). For functional-z the two poles do not overlap. For the empirical p at layer 4 we drew a separate set of 300 random axes, with the two poles chosen separately so that they can share genes.
2. **Strongly co-expressed modules:** each null pole was the set of genes most correlated with a random seed gene (300 axes). These modules are much more co-expressed than the functional poles (Results), so this is a strong null rather than a matched one.
3. **Size- and co-expression-matched sets:** each null pole was built from the k genes most correlated with a random seed gene plus randomly chosen genes, with k chosen by bisection so that the pole’s coherence was within 10% of the functional pole’s (300 axes). We

also kept pole size fixed and traced power against the share of each pole drawn from the genes most co-expressed with the seed gene (0 to 100%, 25 draws per step).

4. **Size- and tightness-matched sets:** the same construction with representation tightness (the mean pairwise cosine of the pole genes' main effects $a(g)$) in place of co-expression (300 axes). When a functional pole was less tight than random genes, so that even a random set was at least as tight, a random set of the same size was used for that pole. This happened in at most 11 of 300 null axes for any axis in the four trained models. In the landmark expression-only version it happened in at most 36 of 300. In the centroid expression-only version it happened in at most 3 of 300. The other pole of each axis was always matched. Such sets make the null harder, not easier, to beat.

For each null, the empirical one-sided p is $(k + 1)/(N + 1)$, where k is the number of the $N = 300$ null axes with power at least as large as the functional axis; when no null axis reached it, p takes its smallest possible value, $1/301$. Empirical p values were used because the null power distributions are right-skewed.

Causal steering test

To test whether the model uses the functional direction, we added it to the residual stream at half of a cell's gene positions and read the model's own next-gene output logits at the other half. The direction was the raw (unstandardised) difference between the mean layer-4 representations of the nuclear-pole and surface-pole genes (poles made disjoint, as for the axes). It was added at the layer-4 representation (the output of the fourth transformer block). For each of 30 immune cells (at most 512 tokens), gene positions were randomly split into a steered half and a read half, so any effect at the read positions must travel through attention. The readout used a wider gene set than the direction: every gene token in MaxToki's vocabulary annotated to a pole (4844

365 nuclear and 5520 surface tokens; genes annotated to both removed). The readout (specificity)
 366 was the mean change in logit over the nuclear-pole tokens minus that over the surface-pole
 367 tokens. Strengths were 0.25, 0.5 and 1.0 times the mean residual-stream norm at that layer in
 368 one reference cell; we report 0.5 as the headline. Pushing toward $+u$ and toward $-u$ and taking
 369 the difference (the signed swing) cancels any effect that does not depend on the sign of the
 370 push. The first control was a random direction of the same norm, drawn fresh for each cell;
 371 significance was a one-sided sign test of functional swing greater than random swing across
 372 cells. This tests whether the push has a consistent effect in the expected direction. Because a
 373 random direction is not built like the axis, a second control was matched to the axis: 30 fixed
 374 random splits of the pooled nuclear-pole and surface-pole genes into two groups of the pole
 375 sizes, each turned into a direction exactly like the functional axis and applied to the same cells,
 376 positions and strength (0.5). The orientation of a random split is arbitrary, so we compared sizes:
 377 $p = (1 + \text{number of splits whose absolute mean swing is at least the absolute functional swing})$
 378 $/ (30 + 1)$, whose smallest possible value is $1/(30 + 1) = 0.032$. We read the direction from
 379 the sign of the functional swing. With 30 splits these p values cannot survive a correction for
 380 several tests, so we report them uncorrected, together with z , the number of standard deviations
 381 by which the functional swing exceeds the absolute split swings. The test was repeated at the
 382 layer-8 representation and for the mitochondrion/cytoskeleton and transcription/transport axes
 383 at layer 4. To put the push sizes in context, we measured how far context naturally moves genes
 384 along the same direction: the spread of each gene's projection onto u across its count-balanced
 385 cell types.

386 To compare MaxToki-217M with MaxToki-1B, we repeated the steering test on the 600-cell
 387 extractions at matched relative depth (MaxToki-217M layer 4, MaxToki-1B layer 7), with a
 388 different push unit. MaxToki-1B has two hidden dimensions that are very large in every gene's

representation. Its raw residual norm is 3.0 times the norm of the part that differs between genes, against 1.1 times in MaxToki-217M, so a push scaled to the raw norm would be much stronger in MaxToki-1B. In this comparison the push unit was therefore the mean norm of the residual after removing its mean over the cell's gene positions (averaged over 5 cells). We ran all three axes in both sizes, with both controls, and repeated the nuclear/surface test in both sizes with the axis, and the split directions, set to zero in the two largest dimensions. The 1000-cell configurations above were also run against the fixed splits.

Tests for context-specific gene function

Directional congruence. Along each functional axis, we decomposed each gene's projection into a gene loading $f(g)$ (mean over contexts), a context loading $h(c)$ (mean over genes) and an interaction $e(g, c)$. If genes move toward their own functional pole specifically in cell types that share that character, e should follow $f \cdot h$. We estimated f and h in one partition and e in the other (and vice versa), measured $\beta = \text{corr}(e, f \cdot h)$ on rank-residualised projections, and compared β with 300 null axes whose poles were matched to the functional poles on size and co-expression (as in null 3 above).

Placement near curated targets. For TFs count-balanced in at least 6 cell types with at least 15 curated targets in the gene panel, we asked (A) whether, in each cell type, the TF's representation (with the context mean removed) is closer by cosine to the centroid of its curated targets than to the centroid of non-targets matched on co-expression with the TF (200 redraws), and (B) whether this excess closeness rises in cell types where the targets are more highly expressed (Spearman correlation across cell types with target activity, the negative of the targets' mean rank). The matched non-targets are matched on co-expression with the TF but not on gene type, and curated target lists are rich in regulators, so (C) we also asked whether the TF is closer to

its own targets than to the targets of the other TFs tested (same cell type, same measure). A cell type was used for a TF only if at least 8 of its targets were count-balanced there, and a TF needed at least 3 such cell types. All three questions were tested across TFs with sign tests.

Lineage transcription factors. We also tried a test on 14 TFs with lineage-specific roles in immune cells (SPI1, IRF4, IRF8, GATA3, MAFB, BCL6, PRDM1, TBX21, RUNX1, RUNX3, BATF, MAF, BACH2 and ZEB2) [40]. It asked whether these TFs move more between cell types than other genes do. A gene's movement was the part of its gene-specific shift in each cell type that repeats in both cell halves (the product of the two halves' shifts, averaged over cell types; MaxToki-217M, 1000-cell extraction, layers 4 and 8). The controls were genes matched on abundance, and genes matched on abundance and on how much their co-expression partners change between cell types; p values came from a permutation that keeps TF families together. Because movement is averaged over cell types, the test needs genes count-balanced in at least 9 of the 12 cell types (Results).

Cross-model, scaling and random-weights comparison

All 600-cell representations (MaxToki-217M, MaxToki-1B, scGPT, STATE, the three untrained models and the three expression-only representations) share the cells, cell types, partitions and gene panel. Model vectors and centroid vectors average 50 occurrences or expressing cells per entry; the landmark and principal-component versions use all 300 cells of each half. EXCESS (on context pairs with at least 150 count-balanced genes) and nuclear/surface functional-z were computed the same way for every representation. Each trained model was scored at two depths (MaxToki-217M layers 2 and 4; MaxToki-1B layers 4 and 7; scGPT layers 2 and 4; STATE layers 3 and 6) and each untrained model at its depth-matched layer. In this comparison functional-z was computed without the rank regression, on genes count-balanced in at least 6 of the 12 cell

types, against 1000 random axes whose two poles do not overlap. In the Setty replication it used genes count-balanced in at least 4 of the 9 clusters and 200 random axes, whose two poles were drawn separately, so they could share genes. Each model has its own count-balanced entries, so we repeated the comparison using only the (cell type, gene) entries that are count-balanced in every representation. The rank regression, all four nulls and both Level 2 tests were also run on the four trained models at their depth-matched layers (MaxToki-217M layer 4, MaxToki-1B layer 7, scGPT layer 4, STATE layer 6) and on the three expression-only representations. For the untrained models we also ran the curated-target test, and for untrained scGPT and STATE the rank regression on the functional axes. On the untrained models only random axes (null 1) were used: we ran none of the co-expression and tightness nulls (nulls 2 to 4) and not the directional-congruence test on them. We built no untrained MaxToki-1B. With 600 cells per cell type fewer genes reach the cap in many cell types, so these runs used genes count-balanced in at least 6 cell types. For the functional axes and the co-expression nulls, at least 9 cell types gave the same conclusions (Additional file 1); the Level 2 tests were run only with at least 6. For the rank regression of non-MaxToki representations we used MaxToki's mean rank on the same cells as an abundance covariate. For the functional axes and the two co-expression nulls we repeated it with the gene's mean $\log_{10}(\text{counts per } 10,000)$ instead.

Link to next-gene prediction

For 350 Tabula Sapiens immune cells with at least 60 expressed vocabulary genes (at most 512 tokens), and up to 20 target positions per cell from the sixth gene onward, we compared the model's log-probability of the true gene given the cell's own preceding genes with that given the same number of preceding genes from another cell (a chimeric context). The donor was another cell, chosen at random, with at least as many genes as the real prefix. The donor genes

used could not include the target gene, so the chimeric context never contains it. The per-gene context benefit (log-probability difference, averaged over occurrences; genes with at least 3 occurrences) was correlated with the gene’s context movement: the mean, over the cell types where the gene is count-balanced (at least 4), of $\|z(g,c) - \text{mean}_c z(g,c)\|$ in the z-scored layer-4 space. This is one number per gene, unlike EXCESS. We used a partial Spearman correlation that controls for the log of the number of times the gene was sampled as a target, with a 95% CI from 2000 bootstrap resamples of genes.

To compare MaxToki-217M and MaxToki-1B, we ran a paired version on the same 350 cells, with each gene’s context movement taken from the 600-cell extractions (MaxToki-217M layer 4, MaxToki-1B layer 7). The cells, target positions and donor cells were drawn from seeds that do not depend on the model, so both models were scored on identical (cell, position, donor) triples. Because MaxToki is causal, the log-probabilities for every prefix of a cell come from one forward pass over the whole cell; on 20 random queries per model this matched separate prefix passes to within 1.2×10^{-5} nats (largest difference). A second donor arm drew the donor from the same cell type, which asks whether context helps beyond cell identity.

Statistics and reproducibility

All random draws use fixed seeds (base seed 0), so every script draws the same cells, genes and null sets each time it runs. Most analyses work from saved hidden states; the steering and prediction-link tests run the model again. Empirical p values are one-sided unless stated. We corrected for multiple tests in three families. Steering (1000-cell MaxToki-217M): at each strength separately, the random-direction sign tests of the four configurations (nuclear/surface at layers 4 and 8; mitochondrion/cytoskeleton and transcription/transport at layer 4), with Bonferroni; the three strengths are not corrected against each other. Curated

481 targets (1000-cell MaxToki-217M): static closeness and its rise with target activity, at lay-
 482 ers 4 and 8, with Bonferroni. Directional congruence: three axes at layers 4 and 8 in the
 483 1000-cell extraction, with Holm, counting only positive β ; the same Holm correction was
 484 applied over each 600-cell representation's own tests (Table 4). The fixed-split p values
 485 are not corrected (see Causal steering test). All other p values are not corrected, including
 486 the null p values (Table 3), the own-versus-other-targets test, the curated-target tests on the
 487 600-cell representations (Table 4) and the 600-cell steering tests. Where Results gives a
 488 corrected value for one of these, it names the correction. Each analysis has its own script:
 489 input encoding `ctx_tokenise.py`; cell selection `ctx_cell_selection.py`; extraction
 490 `ctx_extract_maxtoki.py`, `ctx_extract_scgpt_std.py`, `ctx_extract_state_std.py`,
 491 `ctx_extract_devel.py`, `ctx_extract_random.py` and `ctx_extract_expression.py`;
 492 run configuration for any extraction `ctx_prefix.py`; shared code `ctx_coexpr_null.py`
 493 (only its co-expression matrix function is used; run on its own it gives an older null with
 494 the wrong pole sizes, which `ctx_coexpr_null_v2.py` and `ctx_coexpr_null_v3.py`
 495 replace), `steer_lib.py` (steering and model outputs) and `ctx_stats.py` (gene bootstrap);
 496 `ctx_polysemy.py`, `ctx_layer_curve.py` and `ctx_anisotropy.py` (contextualisation by
 497 layer and anisotropy); `ctx_position_confound.py` (rank control); `ctx_independent.py`
 498 (Setty replication); `ctx_functional_axes.py`, `ctx_celltype_loadings.py`, `ctx_coexpr_null_v2.py`,
 499 `ctx_coexpr_null_v3.py` and `ctx_tightness_null.py` (functional axes and nulls);
 500 `ctx_causal.py` (steering); `ctx_directional_probe.py`, `ctx_curated_targets.py` and
 501 `ctx_switcher_test.py` (tests for context-specific function); `ctx_cross_model.py` (cross-
 502 model comparison); `ctx_context_composition.py` (dataset, donor and assay of each cell
 503 type's cells); `ctx_expression_cellcounts.py` (cells behind each expression-only vector);
 504 `ctx_prediction_link.py` (prediction link); and `ctx_figures.py` (figures). The cell-

selection and extraction scripts save .npz files; the analysis scripts write `results/ctx_*.json`, from which every number and table is taken, and `ctx_figures.py` draws the figures from the same files. The runs used Python 3.12 with PyTorch 2.12.1, transformers 5.13.0 and NumPy 2.4 on Apple-silicon hardware.

Use of large language models

An artificial intelligence assistant (Claude, Anthropic) was used to help write and run the analysis code, check references, and draft and edit the manuscript text. The author designed the study, reviewed the code, results and text, and takes full responsibility for the content. The assistant is not an author.

Results

The first sections use the main MaxToki-217M extraction (1000 cells per cell type) unless a model is named; later sections compare all models on the same 600 cells per cell type. For Level 1, we show that contextualisation is real, is not explained by rank position and replicates on a second dataset, and that the nuclear/surface direction is read by the model when pushed, although not every functional direction is. Two later sections show that contextualisation is partly learned and weakly linked to prediction. For Level 2, we compare the functional organisation with co-expression nulls and with representations built from expression alone, and run two direct tests for context-specific gene function.

Genes are contextualised gene-specifically; in MaxToki-217M most reproducibly in the first layers

The gene-specific context response is large and highly reproducible (Table 1, Fig. 1). At layer 4 of the main extraction, EXCESS is +0.758: the same gene’s relative shift agrees at cosine +0.759 across independent cells, while a different gene’s agrees at +0.001, so the effect is gene-specific. The context main effect replicates at +0.996, which shows that the measurement works. Layer 0 (the context-free embedding output) gives exactly 0.000, as expected: a gene has the same embedding in every cell, so it has no shift to measure. The control that tests the statistic itself is the different-gene agreement (+0.001). The full depth scan shows where the response arises: it is highest at layer 1 (+0.847) and layer 2 (+0.835) and then falls to +0.651 at layer 11 (Fig. 1). The gene-specific shift is therefore already present after the first transformer layer and is most reproducible at layers 1 and 2. EXCESS is a cosine, so it measures how reproducible the direction of the shift is, not how far a gene moves. Deeper layers keep most of it (90% of the peak at layer 4 and 84% at layer 8), while the gene-specific shift becomes somewhat less reproducible toward the output.

Table 1. Gene-specific context response at the four layers of the main MaxToki-217M extraction

Layer	EXCESS (same			Main-effect replication
	– different)	Same gene	Different gene	
0 (embedding)	+0.000	+0.000	+0.000	—
4	+0.758	+0.759	+0.001	+0.996
8	+0.711	+0.711	+0.000	+0.996
11	+0.649	+0.650	+0.001	+0.997

EXCESS is the agreement of the same gene’s relative context shift across two independent halves

of the cells, minus the agreement for a different gene, pooled over 66 cell-type pairs (a median of 1391 count-balanced genes per pair). Gene-level bootstrap 95% CIs are within ± 0.003 . The full depth profile is shown in Fig. 1.

Metric validity (anisotropy). These representations are anisotropic, as expected [9]: the mean cosine of random different-gene pairs is +0.20 to +0.62 at layers 4–11 (raw). After per-dimension z-scoring, which brings anisotropy to about 0.00, same-gene self-similarity across cell types falls with depth: +0.88, +0.65 and +0.61 at layers 4, 8 and 11 (raw, uncorrected: +0.92, +0.85 and +0.69). So genes move more between cell types in upper layers, as Ethayarajh found in language models. Simply subtracting the raw anisotropy baseline gives +0.62, +0.24 and +0.48 instead. This series dips at layer 8, where raw anisotropy peaks, so the layer-8 value mostly reflects the baseline. We z-score each dimension before every cosine-based measure (Methods), because a few dominant dimensions can drive raw cosines and one subtraction does not remove them [13]. With all three versions, self-similarity is higher at layer 4 than at layer 11. Together with the EXCESS profile, this means that upper layers move genes more but less reproducibly. After z-scoring, the first principal component carries only 0.07 of the variance at layer 11, so no single direction dominates the cosines.

The effect is not explained by the gene’s rank position

MaxToki reads a cell as a rank-ordered gene list and encodes token position, so a gene ranking 5th in one cell type and 500th in another is an obvious confound. Three checks show that rank does not drive the gene-specific shift (Fig. 2). At layer 4 the response is about the same for genes whose rank changes less than the median (+0.749; median change 27 positions) and more than the median (+0.767; 107 positions). 99% of it survives a linear regression on the gene’s rank in each cell type and its rank change (+0.749), and 99% survives a regression that also allows

curved effects of rank (+0.747); at layers 8 and 11, 98% and 97% survive the linear regression. And a gene's shift agrees only a little with that of the gene closest to it in rank in both cell types (+0.058, against +0.759 for the same gene and +0.001 for a random gene). Measured against this rank-matched gene instead of a random one, EXCESS is still +0.701 at layer 4 (93% of the full value), and 90% and 88% at layers 8 and 11. So genes with the same ranks share only a small part of the shift. This part is an upper bound on what rank explains, because genes with the same ranks also tend to have similar expression levels. The size of a gene's shift correlates only weakly with its rank change ($\rho = 0.26$ at layer 4 and 0.31 at layer 11). These checks cannot exclude every way in which position could enter a causal model.

The effect replicates on an independent dataset

In MaxToki-217M, on the Setty CD34+ bone-marrow data, with nine differentiation clusters rather than mature cell types as contexts, the phenomenon replicates: EXCESS is +0.713 at layer 2 and +0.713 at layer 4 (gene-shuffled null +0.0036 and +0.0039), and functional-z (without the rank regression, against random axes only; Methods) is +5.0 and +5.0 for the nuclear/surface axis and +16.1 and +8.8 for the mitochondrion/cytoskeleton axis. We did not run the rank control, the co-expression nulls or the Level 2 tests on these data. Using spliced counts only gives the same picture (EXCESS +0.702 and nuclear/surface functional-z +9.3 at layer 4). Neither the contextualisation nor its organisation along these two axes is specific to Tabula Sapiens or to mature cell types.

Context moves genes along functional axes, but not significantly more than along gene sets matched on co-expression

Context moves genes along interpretable functional axes far more than along random directions (Table 2). At layer 4, functional-z (rank-controlled in parentheses) is +16.6 (+13.3) for nuclear versus surface/secreted genes, +8.7 (+6.4) for mitochondrion versus cytoskeleton and +2.9 (+4.3) for transcription versus transport. The cell-cycle versus differentiation axis, which separates its poles only weakly (validity AUC 0.57), gives +2.0 (+0.9). The organisation persists in deeper layers for the two strongest axes. Along the nuclear/surface axis, lymphocytes sit at the nuclear end (CD8 T +4.7, CD4 T +4.4, B +3.9) and alveolar epithelial cells and macrophages at the surface end (alveolar type 1 -3.6, type 2 -3.2, macrophage -2.8). Each cell type is scored on its own set of expressed genes, so this order may mostly show which genes each cell type expresses, not how context moves the same genes (Methods).

Table 2. Functional organisation of context modulation, by axis and layer (MaxToki-217M)

Axis	Poles (genes)	Validity AUC			
		(layer 4)	Layer 4	Layer 8	Layer 11
Nuclear / surface	1800/1188	0.77	+16.6 (+13.3)	+7.8 (+6.5)	+6.1 (+6.0)
Mitochondrion / cytoskeleton	675/165	0.78	+8.7 (+6.4)	+12.7 (+8.4)	+10.4 (+6.9)
Transcription / transport	326/89	0.69	+2.9 (+4.3)	+1.0 (+1.3)	+0.8 (+0.8)

Axis	Poles (genes)	Validity AUC			
		(layer 4)	Layer 4	Layer 8	Layer 11
Cell cycle / differentia- tion	145/167	0.57	+2.0 (+0.9)	+2.9 (+1.7)	+2.6 (+1.9)

Functional-z is the z-score of the axis’s reproducible interaction power against 300 axes built from random gene sets of the same pole sizes, without (and, in parentheses, with) regression on the gene’s rank. Validity AUC is the cross-validated area under the ROC curve with which the axis separates held-out pole genes.

The key control asks whether this organisation exceeds co-expression (Fig. 3). Functional gene sets can be co-regulated, and co-regulated genes co-vary across cells whatever the model “understands”. The functional poles are more co-expressed than random genes, but far less than co-expression modules (mean pairwise correlation 0.028 and 0.042 for the nuclear and surface poles, against 0.025 for random gene sets). At layer 4, against null gene sets matched on size and this average co-expression, no functional axis exceeds the null: the nuclear/surface power (0.50) sits inside the matched null (mean 0.40, 95th percentile 0.68; $p = 0.316$), and $p = 0.375$ and 0.083 for the other two axes. Modules built from the genes most co-expressed with a seed gene, 3–11 times more co-expressed than the functional poles, reach higher power on average (nuclear/surface null mean 0.67; $p = 0.701$, 0.757 and 0.804). We also kept pole size fixed and changed what share of each pole came from the genes most co-expressed with a seed gene. Null power rose steadily with that share. None of the 50 null axes with 0–10% such genes reached the nuclear/surface power (0.50), but 22 of 25 axes made only of such genes did. No single step matches both functional poles. The nuclear pole’s co-expression (0.028) is close to that of

the 10% step (0.027), where 0 of 25 null axes reached the functional power; the surface pole's (0.042) is close to that of the 35% step (0.043), where 10 of 25 did. The matched null, which matches each pole separately, puts the axis inside the null ($p = 0.316$). At layer 4 the organisation is not a by-product of compact gene sets in representation space: it exceeds sets matched on size and representation tightness ($p = 0.007, 0.023$ and 0.013). At layer 8 it does not ($p = 0.233, 0.060$ and 0.100). It far exceeds random axes (none of 300 reached it; $p = 1/301$ for nuclear/surface). So, at layer 4, the functional organisation is not significantly stronger than that of gene sets with the same size and average co-expression. This test can detect only large differences: all three axes sit above the matched-null mean, and an axis would need about 1.7–1.8 times the null mean to reach $p < 0.05$. We ran the co-expression nulls at layer 4 only, not at layers 8 and 11.

Expression alone gives a gene-specific change, but not its organisation

The nulls above ask whether functional gene sets are special among gene sets with the same co-expression, inside the model. A different question is whether a representation built only from expression shows the same contextualisation and organisation. On the same 600 cells per cell type, we compared MaxToki-217M (layer 4) with three expression-only representations (Methods). The landmark and principal-component versions use correlations within one cell type; the centroid version places each gene where it is expressed in a cell space built from all cells pooled. All three change with cell type in a gene-specific way (EXCESS +0.427, +0.218 and +0.321, against +0.738 for MaxToki-217M on the same cells), and 99% of the centroid version's change survives the rank regression. So a reproducible, gene-specific change between cell types does not need a model: part of it follows from where each gene is expressed within each cell type. EXCESS depends on how many cells each vector averages. With all expressing cells instead of 50, the centroid version reaches +0.605, and the two correlation versions use all 300 cells of

each half. These values should therefore not be ranked against the models' (Methods). But none of the three is clearly organised along the functional axes. Rank-controlled functional-z is -1.4 , $+1.8$ and $+1.4$ (landmarks), $+0.6$, $+0.4$ and $+0.8$ (principal components) and -0.3 , -2.1 and $+0.5$ (centroids), against $+16.7$, $+7.5$ and $+5.0$ in MaxToki-217M (Table 3).

The difference also shows in how co-expressed gene sets behave. We express each null's power as a multiple of the power of random poles in the same representation. In MaxToki-217M, sets matched on co-expression reach 1.8–6.0 times random, modules of strongly co-expressed genes 5.6–10.1 times, and poles made only of genes co-expressed with a seed gene 6.1–9.8 times. The other models behave the same way: strong modules reach 5.0–8.9 times random in MaxToki-1B, 3.3–4.2 times in scGPT and 12.5–18.0 times in STATE. The expression-only versions do not. The two within-cell-type versions barely respond to co-expression. Matched sets reach 1.4–1.6 times random with landmarks and 1.3–1.4 times with principal components, and strong modules about the same (1.4–1.5 and 1.5 times). In the centroid version, co-expressed sets do no better than random sets: strong modules reach 0.9–1.0 times random, and poles made only of co-expressed genes 0.9 times. So in every model, context moves genes much more along directions that separate co-expressed gene sets than along random directions, and in none of the three expression-only versions does it. This holds for co-expressed sets in general, not for functional sets in particular. It does not show that the models add information beyond co-expression: other expression-only constructions might come closer (Limitations).

The nuclear/surface direction is read by the model, but not every functional direction is

In the steering test (Fig. 4), adding the nuclear/surface direction to half of a cell's genes moves the model's own logits at the untouched half with the sign of the push. Nuclear-gene logits rise

relative to surface-gene logits for a push toward the nuclear pole (specificity +0.085) and fall relative to them for a push toward the surface pole (−0.097), a signed swing of +0.182. A random direction of the same norm, drawn fresh for each cell, gives +0.001 on average, near zero by design. The swing is +0.079, +0.182, +0.184 at strengths 0.25, 0.5 and 1.0, so it stops growing above 0.5. At 0.5 it exceeds the random swing in 22 of 30 cells ($p = 0.0081$; Bonferroni-adjusted 0.032); at 1.0 in 20 of 30 cells ($p = 0.049$; Bonferroni-adjusted 0.20). The effect is stronger at the layer-8 representation (swing +0.232, 30 of 30 cells, $p = 9.3 \times 10^{-10}$). Against 30 fixed random splits of the same pole genes, built like the axis, the layer-4 swing is larger than 30 of 30 of them (0 of 30 at least as large; $p = 1/31$, $z = +2.6$) and the layer-8 swing larger than 30 of 30 (0 of 30; $p = 1/31$, $z = +6.1$). The direction therefore propagates through attention and changes the model's predictions, more than directions built from random groups of the same genes.

This does not generalise to every functional axis. The mitochondrion/cytoskeleton direction has no significant effect at strengths 0.25 and 0.5 (swing +0.024 and +0.067; above the random swing in 17 of 30 and 15 of 30 cells, $p = 0.29$ and 0.57). At strength 1.0 its swing is +0.269, above the random swing in 20 of 30 cells ($p = 0.049$, Bonferroni-adjusted 0.20). The split control was run only at strength 0.5. There the swing is not distinct from the random splits (18 of 30 splits at least as large; $p = 0.613$). The strength-1.0 swing was not compared with the splits. The transcription/transport direction has a reversed effect that grows with strength (swing −0.139, −0.268, −0.678). At strength 0.5 the functional swing exceeds the random one in only 2 of 30 cells (two-sided sign test $p = 8.7 \times 10^{-7}$; 2 of 30 at 0.25 and 0 of 30 at 1.0). At 0.5 its size also exceeds that of 30 of 30 random splits, but in the reverse direction. We have no explanation for the reversal; it shows that an axis that separates genes is not automatically read in the expected direction. The pushes are also large compared with natural variation. Across cell types, a gene's projection on the nuclear/surface direction has a median range of 2.62 (95th

percentile 4.57). The middle push (7.6 units) is about 3 times that median range, and the largest (15.2 units) about 6 times. The smallest push (3.8 units) is about 1.4 times the median range, below the 95th-percentile range. It already gives a swing of +0.079, above the random direction in 23 of 30 cells ($p = 0.0026$; Bonferroni-adjusted 0.010); it was not tested against the random splits. Even this push moves half of a cell's genes along the axis at once, which natural context does not do. The test therefore shows that the model can read this axis, not that the natural, context-driven changes along it are used.

The nuclear/surface result holds in MaxToki-1B. On the 600-cell extractions at matched relative depth, with the push scaled to the part of the residual that differs between genes, the swing at strength 0.5 is +0.212 in MaxToki-217M and +0.205 in MaxToki-1B. It exceeds a random direction in 25 of 30 and 30 of 30 cells ($p = 1.6 \times 10^{-4}$ and 9.3×10^{-10}). Against 30 random splits, 0 of 30 and 0 of 30 are at least as large (mean absolute split swing 0.067 and 0.047; $p = 1/31$ and $1/31$, $z = +3.6$ and $+4.2$). In MaxToki-1B, 52% of the axis lies in the two very large hidden dimensions. With those coordinates set to zero, the swing is +0.324 (0 of 30 splits at least as large; $z = +6.3$), so they do not carry the effect. In MaxToki-217M, which has no such very large dimensions, only 13% of the axis lies in its own two largest dimensions. Even so, with them set to zero the swing falls from +0.212 to +0.131 (splits at least as large: 1 of 30; $p = 0.065$, $z = +1.9$), so the swing is no longer clearly above the splits. In this model part of the effect goes through its two largest dimensions. The other two axes behave differently in the two sizes. In MaxToki-1B both are read in the expected direction: the swing is +0.400 for mitochondrion/cytoskeleton ($z = +4.7$ against the splits) and +0.111 for transcription/transport ($z = +4.6$). In MaxToki-217M the transcription/transport direction is again read in reverse (-0.242), and the mitochondrion/cytoskeleton swing (+0.168) is not distinct from the random splits (7 of 30 at least as large; $p = 0.258$). So only the nuclear/surface direction is read as expected in both

sizes and both extractions; in the larger model all three directions are.

No context-specific gene function beyond co-expression

Two direct tests for context-appropriate gene function find no context-specific signal beyond co-expression. (1) Directional congruence asks whether genes move toward their own functional pole in the cell types that lean toward that pole (β , Methods). For no axis was β significantly larger than for null axes matched on size and co-expression. At layer 4, β is +0.16, +0.10 and +0.23 for the three axes, all positive but within the upper tail of size- and co-expression-matched null axes, whose means are near zero ($p = 0.116, 0.086$ and 0.053 ; none significant after Holm correction over 6 tests; at layer 8, β is $-0.16, -0.11$ and $+0.09$). So at layer 4 the trend points the expected way but is not significant; the null spread of β (standard deviation 0.08 – 0.13) means the test could detect only clear congruence. (2) Placement near curated targets: across 26 TFs, at layer 4 a TF's representation is slightly closer to the centroid of its curated targets than to that of co-expression-matched non-targets (mean cosine excess $+0.021$; 20 of 26 TFs positive; sign test $p = 0.005$, Bonferroni-adjusted over 4 tests 0.019). Compared with the targets of the other TFs tested, a TF is only slightly closer to its own targets ($+0.008$; 17 of 26 TFs positive; $p = 0.084$), so the closeness is not shown to be specific to each TF's own targets; much of it may reflect where regulators and their usual targets sit in general. The closeness does not rise significantly in the cell types where the targets are more active (mean $\rho = +0.084$; 15 of 26 TFs positive; $p = 0.28$, Bonferroni-adjusted 1.00); the trend is positive but weak, and each TF's ρ rests on as few as 3 cell types. At layer 8 neither effect is significant (static $p = 0.423$; modulation $\rho = +0.097$, $p = 0.28$). We did not run an untrained model on these cells. On the 600-cell extraction only 11 TFs qualify (Table 4). There the closeness is smaller and not significant ($+0.011$; 8 of 11 TFs positive; $p = 0.113$). The untrained MaxToki-217M gives a mean no smaller ($+0.030$), but only

6 of 11 TFs are positive ($p = 0.500$). That 11-TF test has little power, so we cannot say whether the closeness is learned, or whether it holds beyond these 26 TFs. A third attempt, on 14 lineage TFs, is uninformative rather than negative. The test needs a gene to be count-balanced in at least 9 of the 12 cell types. 10 of the 14 TFs are in the 6000-gene panel, but only 1 (RUNX1) met that bar. These regulators are weakly expressed, or are expressed in only a few of the cell types we used.

MaxToki-1B, scGPT and STATE also contextualise genes, but show no context-specific gene function beyond co-expression

We ran the full set of controls and both Level 2 tests on every trained model and expression-only representation of the same 600 cells per cell type (Table 3, Table 4). Three results hold in every model ((1), (3) and (4) below), one holds only in some models ((2)), and one holds only in MaxToki-217M ((5)). (1) Every model contextualises genes strongly and gene-specifically (EXCESS +0.738, +0.764, +0.813 and +0.867 for MaxToki-217M, MaxToki-1B, scGPT and STATE), and almost all of it survives the rank regression (99%, 99%, 98% and 99%). (2) Organisation beyond random axes holds only partly. The MaxToki models organise the change along all three functional axes beyond random axes (rank-controlled functional-z for the nuclear/surface axis +16.7 and +8.0), whichever abundance covariate is used (+16.8 and +7.8 with mean expression). STATE does so too (+5.5, +9.8 and +4.0). But when the gene's mean expression is the covariate, STATE's nuclear/surface organisation disappears (+0.2). Its other two axes remain (+7.0 and +3.3). STATE and scGPT receive each gene's expression level as input; MaxToki receives only the genes' rank order. scGPT organises the change only weakly (+1.4, +2.3 and +3.5). (3) In no representation does any functional axis significantly exceed gene sets matched on size and co-expression (smallest $p = 0.07, 0.19, 0.23$ and 0.32), or modules of

751 strongly co-expressed genes. (4) No model shows context-specific gene function. Directional
 752 congruence is not significant after correction in any of the four models, and in none does a TF's
 753 closeness to its curated targets rise in the cell types where those targets are active. Expression
 754 alone can pass, or nearly pass, each test. In the centroid version we use, the closeness rises
 755 with target activity (mean $\rho = +0.214$; 9 of 11 TFs positive; $p = 0.03$, not corrected; 0.10 after
 756 Bonferroni over the three curated-target questions). Its build with all expressing cells, which we
 757 report only as a check, passes the directional test on the transcription/transport axis ($\beta = +0.11$,
 758 $p = 0.007$, significant after Holm correction over 3 tests), where its matched null is narrow (stan-
 759 dard deviation 0.05, against 0.11–0.16 in MaxToki-217M); the build we use does not ($p = 0.113$).
 760 So a pass on either test would not by itself show gene function beyond expression. One static
 761 signal appears: STATE places TFs closer to their curated targets than to co-expression-matched
 762 non-targets (24 TFs, mean cosine excess $+0.099$, $p = 7.7 \times 10^{-4}$). Compared with the targets of
 763 the other TFs tested, a STATE TF is slightly closer to its own targets ($+0.047$; 17 of 24 TFs pos-
 764 itive; $p = 0.032$, not corrected; 0.096 after Bonferroni over the three curated-target questions).
 765 So we cannot say that the closeness is specific to each TF's own targets. The untrained STATE,
 766 which uses the same fixed protein-embedding gene table as the trained model, shows the same
 767 closeness ($+0.132$, $p = 1.4 \times 10^{-4}$) and is about as specific to each TF's own targets ($+0.040$; 17
 768 of 24 TFs positive; $p = 0.032$). Untrained scGPT and MaxToki-217M show no significant close-
 769 ness (-0.008 , $p = 0.895$; $+0.030$, with 6 of 11 TFs positive, $p = 0.500$; this 11-TF test has little
 770 power). So STATE's closeness does not come from training on expression. It most likely comes
 771 from the protein-sequence embeddings; we did not run STATE with a random gene table, which
 772 would test this directly. scGPT shows a different pattern. Its TFs are not closer to their targets
 773 than to co-expression-matched non-targets ($+0.026$; $p = 0.402$). But they are closer to their own
 774 targets than to the targets of the other TFs tested ($+0.070$; 13 of 16 TFs positive; $p = 0.011$, not

corrected; $p = 0.011$ at layer 2), and untrained scGPT is not (-0.005 ; $p = 0.962$). This comparison is not matched on co-expression, so the scGPT result most likely reflects co-expression between a TF and its targets, learned in training, rather than a signal beyond co-expression. (5) Only MaxToki-217M passes the tightness control. It clearly exceeds gene sets matched on size and representation tightness on two of its three axes ($p = 1/301$ for nuclear/surface and $1/301$ for transcription/transport; mitochondrion/cytoskeleton $p = 0.053$). MaxToki-1B reaches $p = 0.033$ on one axis (transcription/transport); this p is not corrected and does not survive a Holm correction over the model's three axes (0.10). scGPT and STATE exceed the null on no axis (Table 3). So only in MaxToki-217M is the organisation clearly not a by-product of compact gene sets.

Table 3. Functional organisation and its controls in every representation, on the same cells

Representation	EXCESS	Kept	Functional-		p,	p, strong	
		after rank	z,	p,	matched	co-	p,
		regres-	rank-	random	co-	expression	matched
		sion	controlled	axes	expression	modules	tightness
MaxToki-	+0.738	99%	+16.7 /	1/301 /	0.35 /	0.78 /	1/301 /
217M			+7.5 /	1/301 /	0.46 /	0.73 /	0.053 /
(layer 4)			+5.0	0.007	0.07	0.75	1/301
MaxToki-	+0.764	99%	+8.0 /	1/301 /	0.75 /	0.95 /	0.385 /
1B (layer			+8.1 /	1/301 /	0.39 /	0.61 /	0.824 /
7)			+3.1	0.017	0.19	0.80	0.033
scGPT	+0.813	98%	+1.4 /	0.146 /	0.91 /	0.96 /	0.674 /
(layer 4)			+2.3 /	0.037 /	0.53 /	0.72 /	0.316 /
			+3.5	0.017	0.23	0.45	0.146

		Kept	Functional-		p,	p, strong	
		after rank	z,	p,	matched	co-	p,
		regres-	rank-	random	co-	expression	matched
Representation	EXCESS	sion	controlled	axes	expression	modules	tightness
STATE	+0.867	99%	+5.5 /	0.010 /	0.71 /	0.95 /	0.369 /
(layer 6)			+9.8 /	1/301 /	0.36 /	0.64 /	0.143 /
			+4.0	0.017	0.32	0.82	0.289
Expression	+0.427	96%	-1.4 /	0.934 /	0.98 /	0.99 /	0.997 /
only,			+1.8 /	0.027 /	0.29 /	0.37 /	0.352 /
land-			+1.4	0.100	0.39	0.42	0.389
marks							
Expression	+0.218	96%	+0.6 /	0.163 /	0.42 /	0.62 /	0.741 /
only,			+0.4 /	0.269 /	0.57 /	0.73 /	0.837 /
principal			+0.8	0.269	0.50	0.64	0.605
compo-							
nents							
Expression	+0.321	99%	-0.3 /	0.528 /	0.33 /	0.42 /	0.654 /
only,			-2.1 /	0.993 /	1.00 /	0.98 /	0.990 /
centroids			+0.5	0.312	0.23	0.22	0.365

Each cell lists the nuclear/surface, mitochondrion/cytoskeleton and transcription/transport axes, in that order. All representations use the same 600 cells per cell type, partitions and gene panel; genes count-balanced in at least 6 of the 12 cell types. The p values are empirical one-sided values against 300 null axes of the same pole sizes (1/301 is the smallest possible). Rank regres-

789 sion uses MaxToki's mean rank on the same cells as the abundance covariate. Additional file 1
790 repeats the functional axes and the co-expression nulls with the mean log-normalised expression
791 as covariate, and with genes count-balanced in at least 9 cell types.

Table 4. Tests for context-specific gene function in every representation

Representation	Directional congruence β	p (smallest at this layer; Holm- significant tests, all layers)		TFs tested	Closeness to curated targets (p)	Closeness to other TFs’ targets (p)	Closeness to other TFs’ targets (p)	Rise where targets are active, ρ (p)
MaxToki- 217M (layer 4)	+0.12 / −0.07 / +0.15	0.086 (0)	11		+0.011 (0.113)	+0.004 (0.887)		+0.028 (0.89)
MaxToki- 1B (layer 7)	+0.07 / −0.20 / +0.02	0.306 (0)	11		+0.014 (0.274)	+0.001 (0.726)		+0.068 (0.50)
scGPT (layer 4)	−0.05 / −0.02 / −0.16	0.482 (0)	16		+0.026 (0.402)	+0.070 (0.011)		−0.002 (0.40)
STATE (layer 6)	−0.03 / +0.09 / −0.08	0.266 (0)	24		+0.099 (7.7×10^{-4})	+0.047 (0.032)		−0.047 (0.73)

		p (smallest			Closer to	
		at this			own than	Rise where
		layer;				
		Holm-				
	Directional	significant		Closeness	to other	targets are
	congru-	tests, all		to curated	TFs'	active, p
Representation	nance β	layers)	TFs tested	targets (p)	targets (p)	(p)
Expression	-0.03 /	0.106 (0)	11	-0.098	-0.007	+0.037
only,	-0.37 /			(0.887)	(0.726)	(0.73)
landmarks	+0.43					
Expression	+0.04 /	0.206 (0)	11	-0.029	+0.014	+0.131
only,	+0.14 /			(0.887)	(0.500)	(0.50)
principal	+0.25					
compo-						
nents						
Expression	+0.00 /	0.113 (0)	11	+0.049	+0.076	+0.214
only,	-0.11 /			(0.500)	(0.500)	(0.03)
centroids	+0.05					

Directional congruence β is listed for the nuclear/surface, mitochondrion/cytoskeleton and tran-
 scription/transport axes, against 300 size- and co-expression-matched null axes; the Holm cor-
 rection runs over all axes and layers tested for that representation (6 tests for the MaxToki models,
 9 for scGPT, 9 for STATE and 3 for the expression-only representations). The curated-target
 tests use TFs count-balanced in at least 6 cell types with at least 15 curated targets in the panel;

with 600 cells per cell type fewer TFs qualify than in the 1000-cell MaxToki-217M extraction (26 TFs): 11 for the MaxToki models and the expression-only representations, 16 for scGPT and 24 for STATE. The 11-TF tests have little power. The third test compares a TF's closeness to its own targets with its closeness to the targets of the other TFs tested. The curated-target p values are not corrected for multiple tests.

Contextualisation across architectures and sizes, and with untrained models

Functional organisation can be measured whether or not a model shows context-specific gene function, so it could serve as a way to compare models (Table 5, Fig. 5). Here every representation shares the cells, cell types, partitions and gene panel. (1) Every model contextualises genes strongly (EXCESS +0.740, +0.767, +0.813 and +0.867 for MaxToki-217M, MaxToki-1B, scGPT and STATE at depth-matched layers; these differ slightly from Table 3 because more cell-type pairs are scored here, Methods), but contextualisation does not grow consistently with model size. At matched relative depth, MaxToki-1B gives +0.767 (layer 7) against +0.740 for MaxToki-217M (layer 4), and +0.766 (layer 4) against +0.832 (layer 2). Functional organisation is not consistently higher in the larger model either: functional-z is +7.2 against +15.5 at relative depth 0.35, and +18.0 against +18.4 at 0.2. The depth pattern also differs between models. From the earlier to the later layer, EXCESS falls in MaxToki-217M (+0.829 to +0.738) and slightly in STATE (+0.879, +0.867 and +0.842 at layers 3, 6 and 12), stays the same in MaxToki-1B (+0.764 and +0.764) and rises in scGPT (+0.687, +0.813 and +0.872 at layers 2, 4 and 9; all depth values in this sentence use cell-type pairs with at least 200 count-balanced genes, as in Table 3). (2) On the nuclear/surface axis and each model's own entries, functional organisation is highest in the MaxToki models and much lower in scGPT and STATE (+3.7 and +4.0). The order

820 between the other models depends on the gene set and the controls. On the entries shared by all
 821 representations, STATE (+6.8) is above scGPT (+3.3) and MaxToki-1B at layer 7 (+4.8). After
 822 the rank regression, STATE is above both MaxToki models on the mitochondrion/cytoskeleton
 823 axis (+9.8 against +7.5 and +8.1; Table 3), but with the gene's mean expression as the covariate
 824 it is below both (+7.0 against +9.1 and +8.8). On the nuclear/surface axis, only MaxToki-217M
 825 stays clearly above scGPT and STATE in every version: each model's own entries, the shared
 826 entries, the 9-cell-type cut and both abundance covariates. On the other two axes it is not clearly
 827 above STATE. The models differ in more than input (training data, objective and gene vocabulary),
 828 so this comparison does not isolate the cause, and the functional organisation is not yet a
 829 stable way to rank architectures. (3) Part of the contextualisation needs no training in every architecture.
 830 Untrained versions keep 60% (MaxToki-217M), 36% (scGPT) and 88% (STATE) of
 831 the trained EXCESS. Untrained MaxToki-217M keeps a much smaller share of the functional organisation
 832 (functional-z +4.6 against +15.5; +4.1 against +11.3 on the shared entries). In scGPT
 833 and STATE the trained values are already low (+3.7 and +4.0), so these comparisons tell us little
 834 about whether training adds organisation. Untrained scGPT gives +2.2. Untrained STATE
 835 gives +1.4 on its own entries, but +3.8 against +6.8 on the shared entries. After the rank regression,
 836 untrained STATE keeps about half of the trained organisation on the nuclear/surface and
 837 mitochondrion/cytoskeleton axes (+3.0 against +5.5; +4.8 against +9.8) and is above the trained
 838 model on transcription/transport (+5.6 against +4.0). So in STATE much of the functional organisation
 839 does not need training. In untrained scGPT the functional axes barely separate their
 840 two poles (validity AUC 0.51–0.57), so its values cannot be read. We did not test why the untrained
 841 STATE keeps most of the contextualisation. Two things could contribute: each gene's
 842 input token carries its expression level, which itself differs between cell types, and the untrained
 843 model uses the same fixed protein-embedding gene table as the trained one. Running STATE

with random weights and a random gene table would separate the two; we did not run it. The expression-only representations show little or no functional organisation (-1.3 , $+0.7$ and -0.4). Their EXCESS is lower ($+0.424$, $+0.218$ and $+0.320$), but it depends on how many cells each vector averages ($+0.607$ for the centroid version with all expressing cells), so the strength of contextualisation alone does not separate the models from expression. On the 14,109 shared entries EXCESS changes little. Besides the changes given above, on the shared entries MaxToki-1B is above MaxToki-217M at relative depth 0.2 ($+15.1$ against $+13.5$), and untrained STATE is above trained scGPT ($+3.8$ against $+3.3$).

Table 5. Contextualisation strength and functional organisation of every representation on the same cells

Representation	Layer		Genes	EXCESS	z	EXCESS,	Functional-
	(relative depth)	Width				shared entries	z, shared entries
MaxToki-217M	2 (0.18)	1232	4425	+0.832	+18.4	+0.833	+13.5
MaxToki-217M	4 (0.36)	1232	4425	+0.740	+15.5	+0.735	+11.3
MaxToki-1B	4 (0.20)	2304	4425	+0.766	+18.0	+0.763	+15.1
MaxToki-1B	7 (0.35)	2304	4425	+0.767	+7.2	+0.761	+4.8
scGPT	2 (0.17)	512	4093	+0.688	+4.0	+0.650	+4.6
scGPT	4 (0.33)	512	4093	+0.813	+3.7	+0.800	+3.3

Layer		EXCESS, Functional-					
(relative		Functional- shared				z, shared	
Representation	depth)	Width	Genes	EXCESS	z	entries	entries
STATE	3 (0.19)	2048	4761	+0.880	+4.0	+0.863	+7.9
(SE-							
600M)							
STATE	6 (0.38)	2048	4761	+0.867	+4.0	+0.853	+6.8
(SE-							
600M)							
MaxToki-	4 (0.36)	1232	4425	+0.441	+4.6	+0.443	+4.1
217M,							
untrained							
scGPT,	4 (0.33)	512	4093	+0.296	+2.2	+0.238	−0.7
untrained							
STATE,	6 (0.38)	2048	4761	+0.761	+1.4	+0.745	+3.8
untrained							
Expression	—	512	4425	+0.424	−1.3	+0.447	−1.3
only,							
land-							
marks							

Layer		EXCESS, Functional-				
(relative		Functional- shared			z, shared	
Representation	depth)	Width	Genes	EXCESS	z	entries
Expression —		256	4425	+0.218	+0.7	+0.227
only,						+0.3
principal						
compo-						
nents						
Expression —		50	4425	+0.320	−0.4	+0.328
only,						−0.6
centroids						

852 All representations use the same 600 cells per cell type, the same two partitions and the same
 853 5000-gene panel. Model vectors average 50 occurrences and centroid vectors 50 expressing
 854 cells per entry; the landmark and principal-component versions use all 300 cells of each half.
 855 EXCESS measures how reproducible the gene-specific context shift is; functional-z measures or-
 856 ganisation along the nuclear/surface axis against 1000 random-gene axes with non-overlapping
 857 poles (no rank regression; genes count-balanced in at least 6 of the 12 cell types). A cell-type
 858 pair is scored if it has at least 150 count-balanced genes. This keeps all 66 pairs in the per-
 859 model columns and 64 pairs in the last two columns, where only shared entries are used. Table
 860 3 uses at least 200, so EXCESS there can differ by up to 0.003. “Genes” is the number of panel
 861 genes count-balanced in at least one cell type; each representation has its own count-balanced
 862 entries. The last two columns repeat the analysis on the 14,109 (cell type, gene) entries, from
 863 3957 genes, that are count-balanced in every representation. Relative depth is the number of

transformer blocks applied divided by the model's number of blocks. EXCESS values of different architectures should not be ranked against each other, because width and input type differ (Limitations).

Contextualisation is weakly correlated with what the model gains from context in prediction

If contextualisation serves the autoregressive objective, genes the model contextualises more should gain more from real context in next-gene prediction. We find a weak positive link of this kind. Real context helps prediction a great deal: across 2266 comparisons for 558 genes, the mean benefit is +12.0 nats (median +12.9; a nat is a natural-log unit of probability), so the true gene is roughly 160,000-fold more probable under its real context than under another cell's. The per-gene size of this benefit is weakly related to how much the gene moves between cell types: the partial Spearman correlation, controlling for how often the gene was sampled as a target, is +0.161 (95% CI +0.083 to +0.239). Genes whose representations depend more on context tend, weakly, to be genes whose prediction depends more on context.

The link holds in MaxToki-1B and beyond cell identity. In the paired design (the same 350 immune cells, positions and donors for both models, with each gene's movement taken from the 600-cell extractions), the partial correlation is +0.133 (95% CI +0.040 to +0.222) for MaxToki-217M and +0.200 (+0.108 to +0.280) for MaxToki-1B with a random donor. When the donor comes from the same cell type, real context still helps a great deal (mean benefit +10.83 and +10.36 nats, against +12.07 and +11.53 with a random donor), and the correlation stays positive (+0.236, CI +0.147 to +0.322; +0.323, CI +0.242 to +0.400). It is higher than with a random donor in both models, but we did not test this difference, and the two arms use slightly different genes. So most of the benefit of real context is specific to the cell, not to its cell type, and it is

somewhat larger for the genes the model contextualises more. These tests used immune cells only.

Discussion

The results characterise how SCFMs represent genes in context and what that representation carries. All four models change a gene’s representation with cell type in a gene-specific and reproducible way. In MaxToki-217M, the only model we measured at every layer, this change is most reproducible in the first one or two layers, while genes keep moving between cell types in later layers. This depth pattern is not shared by every model. The other models were measured at only two or three layers: in scGPT, EXCESS rises with depth (+0.687, +0.813 and +0.872 at layers 2, 4 and 9), in STATE it falls slightly (+0.879, +0.867 and +0.842 at layers 3, 6 and 12), and in MaxToki-1B it is the same at layers 4 and 7 (+0.764). Untrained versions of the three architectures keep part of the change (36–88% of the trained EXCESS), so some contextualisation comes from the architecture and the input alone. Training strengthens it. In MaxToki-217M it also strengthens its alignment with functional axes (functional-z +4.6 before training and +15.5 after); in STATE much of that alignment is already there before training. In both MaxToki sizes the change is weakly correlated, gene by gene, with how much the model’s next-gene predictions in immune cells depend on the cell’s own context, beyond its cell type. To our knowledge this is the first measurement for single-cell models of the “how contextual are contextualised representations” question [9]. On the nuclear/surface axis, the functional organisation of the change separates MaxToki-217M from scGPT and STATE in every version of the analysis (functional-z +15.5 against +3.7 and +4.0; +16.7 against +1.4 and +5.5 after the rank regression). On the other two axes it does not: on the mitochondrion/cytoskeleton axis STATE is higher after the rank regression (+9.8 against +7.5) but lower with mean expression as the

910 covariate (+7.0 against +9.1), and on the transcription/transport axis the three models are close
911 (+5.0, +3.5 and +4.0). The measure also does not order the other models consistently, so it is
912 not yet a stable way to rank architectures.

913 What the change carries is organised by co-expression. In no model is the functional organi-
914 sation significantly stronger than that of gene sets matched to the functional sets on size and
915 average co-expression (smallest $p = 0.07$), and modules of strongly co-expressed genes pro-
916 duce more. This test can detect only large differences, so a small functional effect beyond
917 co-expression cannot be ruled out. Expression alone gives part of this picture but not all of it. A
918 representation that places each gene where it is expressed within each cell type also changes in a
919 gene-specific way (EXCESS +0.321 when each vector averages 50 cells, as in the models, and
920 +0.605 with all expressing cells), so a reproducible, gene-specific change is not by itself a sign
921 of learning. EXCESS values of representations with different widths, inputs and sample sizes
922 should not be ranked against each other. But in none of the three expression-only versions is the
923 change organised along functional axes or along co-expressed gene sets: strong co-expression
924 modules reach 1.4–1.5, 1.5 and 0.9–1.0 times the power of random gene sets, against 5.6–10.1
925 times in MaxToki-217M, 5.0–8.9 in MaxToki-1B, 3.3–4.2 in scGPT and 12.5–18.0 in STATE.
926 So the models move genes along directions that separate co-expressed gene sets much more
927 than these expression-only versions do, for co-expressed sets in general rather than for func-
928 tional sets in particular. We do not claim that the models add information beyond co-expression;
929 other expression-only constructions, in particular ones that see co-expression between cell types,
930 might come closer. The two direct tests for context-specific gene function are negative in every
931 model. An expression-only representation can pass the directional test (the centroid build with
932 all expressing cells) and nearly pass the target-activity test (the centroid build we use; $p = 0.03$
933 before correction), so a pass would not by itself show function beyond expression. The one

934 signal beyond co-expression is static: in STATE, and more weakly in the larger MaxToki-217M
935 extraction, TFs sit closer to their curated targets than to co-expression-matched non-targets, but
936 not more so in the cell types where those targets are active. Compared with the targets of other
937 TFs, the closeness is not clearly specific to each TF's own targets in either model (STATE: 17
938 of 24 TFs positive, $p = 0.032$ before correction; MaxToki-217M: 17 of 26 TFs, $p = 0.084$),
939 so it may partly reflect where regulators and their usual targets sit in general. scGPT TFs are
940 closer to their own targets than to other TFs' targets, but not closer than co-expression-matched
941 non-targets, so that result is consistent with co-expression alone. In STATE this closeness is
942 already present before training and, if anything, larger; untrained models without a fixed gene
943 table show none. So it most likely comes from the protein-sequence embeddings that serve as
944 STATE's gene tokens, not from learning on expression data; we did not run STATE with a ran-
945 dom gene table, which would test this directly. For MaxToki-217M we cannot say whether the
946 closeness is learned: we did not run an untrained model on the 1000-cell extraction, and on the
947 600-cell extraction neither the trained nor the untrained model shows it.

948 The steering results do not test co-expression. Both MaxToki sizes read the nuclear/surface di-
949 rection more than directions built in the same way from random splits of the same genes. In
950 MaxToki-217M part of this effect goes through its two largest hidden dimensions: with them
951 set to zero, the effect is no longer clearly above the splits ($p = 0.065$). MaxToki-1B also reads the
952 other two functional directions. The random splits mix the two poles, so they are not matched
953 to them on co-expression, and a model that reads co-expressed gene sets would also pass this
954 test. The pushes are about 1.4 to 6 times a gene's typical (median) natural range along the axis,
955 and each moves half of a cell's genes at once, so the tests show that the models can read these
956 directions, not that natural context changes along them are used. MaxToki-217M reads the tran-
957 scription/transport direction in reverse, and its mitochondrion/cytoskeleton direction is not dis-

958 tinct from random splits. The architecture comparison does not contradict the main conclusion.
959 On the nuclear/surface axis STATE organises the change more weakly than MaxToki-217M in
960 every version we ran, and its organisation there disappears when the gene’s mean expression is
961 the covariate (+0.2, against +5.5 with the rank covariate). Against MaxToki-1B the order de-
962 pends on the gene set: on each model’s own entries STATE is lower (+5.5 against +8.0 after the
963 rank regression), and on the entries shared by all representations it is higher (+6.8 against +4.8).
964 On the other two axes STATE is about as strong as the MaxToki models. scGPT organises the
965 change only weakly. In all four models the organisation stays within the co-expression nulls.
966 The comparison cannot separate input type from the other differences between the models.

967 One possible reason why we found no context-specific gene function beyond co-expression is
968 that these models are trained only on expression data. They see which genes are expressed and
969 either their rank order (MaxToki) or their expression level (scGPT and STATE). STATE also
970 gets a fixed protein-sequence embedding for each gene. Their training rewards predicting co-
971 expressed genes. The link between contextualisation and the benefit of real context is consistent
972 with that reading, although the test measures only how far a gene moves, not in which direc-
973 tion. Some context-dependent function acts at the protein level, as in moonlighting proteins that
974 switch function with location or binding partners [39], and need not change expression. Other
975 forms, such as cell-type-specific enhancer choice by lineage-determining TFs [40], act through
976 target-gene expression and could in principle be visible; our tests did not detect them. STATE’s
977 protein-sequence gene tokens are the same in every cell type, so they cannot carry a gene’s
978 changed role by themselves. We did not test multi-omic models, models that read regulatory
979 DNA sequence together with expression, or much larger expression models.

Limitations

The depth of analysis differs between models. The rank control, the co-expression and tightness nulls, both Level 2 tests and the expression-only comparison were run on the four trained models and three expression-only representations, on the same 600 cells per cell type. The untrained models went through EXCESS, functional-z against random axes and the curated-target test; untrained scGPT and STATE also went through the rank regression on the functional axes. Each untrained model was built once from one random seed, and we built an untrained MaxToki-217M only, not an untrained MaxToki-1B. The Setty replication, the full depth profile and the 1000-cell analyses were run on MaxToki-217M only; on the Setty data we computed only EXCESS and functional-z against random axes. Steering and the prediction link were run on the two MaxToki models only, and the cells that were pushed or scored were immune cells (30 for steering, 350 for the prediction link). We did not steer scGPT or STATE: neither has a next-gene readout comparable to MaxToki's, and their expression heads would first need a check that they track the input, which we did not do. With 600 cells per cell type fewer genes reach the occurrence cap in many cell types, and only 11 TFs qualified for the curated-target test in the MaxToki models and the expression-only representations (16 for scGPT, 24 for STATE), so that test has little power for those representations. The Level 2 tests were run only with genes count-balanced in at least 6 cell types. Each model has its own count-balanced entries; on the entries shared by all representations EXCESS barely changes, but the functional-z order changes for STATE, scGPT and MaxToki-1B, and only EXCESS and nuclear/surface functional-z were repeated on the shared entries. Functional-z also moves with the draw of random axes, so only clear differences between models should be read as real. EXCESS values should not be ranked across architectures, because widths and input types differ. Input lengths also differ: MaxToki saw each cell's top 1022 genes by count divided by gene median, scGPT a random 1199 of its

1004 expressed genes and STATE the top 2047 by count. scGPT and STATE receive each gene's
1005 expression level; the rank regression used MaxToki's mean rank on the same cells. Repeating it
1006 with mean log-normalised expression left every co-expression-null result and MaxToki's organ-
1007 isation unchanged but removed STATE's nuclear/surface organisation (Additional file 1); we
1008 did not repeat it for the EXCESS rank regression, the tightness null or the two Level 2 tests.

1009 The matched co-expression null is wide: depending on the model, an axis would need about 1.5
1010 to 2.9 times the matched-set mean to reach $p < 0.05$, so this test could detect only a large excess.
1011 It also matches only average pairwise co-expression; power depends on how co-expression is
1012 distributed within a gene set. Two of the three expression-only representations use correlations
1013 within each cell type and barely respond to co-expression. The third is built on axes from all
1014 cell types pooled, but each vector is still an average over cells of one cell type, so it may not
1015 capture co-expression between cell types either; none of the three responds to co-expressed
1016 gene sets. So the contrast between the models and these versions does not show that the models
1017 add anything beyond co-expression in the data. The third version's result also depends on how
1018 it is built: with 512 instead of 50 principal components it failed the positive control, and we
1019 did not run other numbers of components through the pipeline. Other constructions, such as
1020 gene embeddings learned from the same cells or a co-expression network embedding, were not
1021 tested and might reproduce more of the organisation. The rank control rests on regressions
1022 and a rank-matched comparison; it cannot exclude every way in which position could enter a
1023 causal model. Most analyses use one layer per model, at matched relative depth, not the peak
1024 layers 1–2 of MaxToki-217M; the other models were not measured at every layer. The steering
1025 direction is the context-averaged difference between two gene classes, not the context-specific
1026 change itself, and the pushes are large; we did not steer along directions built from gene sets
1027 matched to the poles on co-expression, so steering cannot separate function from co-expression;

1028 and with 30 random splits, the split-control p values cannot survive a correction for several tests.
1029 The prediction link is a correlation across genes, and it controls only for how often a gene was
1030 sampled as a target. A gene whose expression varies more from cell to cell could both move
1031 more between cell types and gain more from its own context, whatever the model does with that
1032 movement. We did not control for this, for example with the gene's rank change or the spread of
1033 its expression, and we did not check whether movement measured in the untrained models or the
1034 expression-only representations shows the same link. The Setty data cover a pre-filtered set of
1035 genes, unlike the full raw counts of Tabula Sapiens. Cell type is partly tied to tissue and donor.
1036 Of the 12 cell types, 5 come from one dataset only: 4 from lung, and kidney epithelial cells,
1037 all from a single donor. Most pulmonary alveolar type 1 cells (915 of 1000) also come from
1038 one donor. Assay mix differs less: in every cell type at least 86% of cells are 10x 3' v3. The
1039 two halves of a cell type share its tissue, donor and ambient-RNA background, so differences in
1040 these between cell types count toward EXCESS and repeat in both halves. We did not measure
1041 how large this part is, for example by comparing cell types from the same tissue and donor. In
1042 the Setty data all contexts come from one bone-marrow dataset and EXCESS is similar (+0.713),
1043 so the effect does not need these differences, but their share in the Tabula Sapiens numbers is
1044 unknown. The Level 2 results are negative, but they do not show that context-specific gene
1045 function does not exist. Some of it leaves no trace in expression, and the directional test could
1046 detect only clear congruence, so weaker effects could be missed. Functional axes are built from
1047 direct annotations in one annotation file; the conclusions hold across three axes, but specific
1048 z-scores depend on how the poles are defined.

Conclusions

The four models we tested build contextual gene representations that are worth studying in their own right. A gene's vector changes with its cell type in a gene-specific, reproducible way. Part of this change comes from the architecture and part from training, and a representation built from expression alone also changes in a gene-specific way. In MaxToki-217M, the only model scanned layer by layer, it is most reproducible in the first one or two layers; scGPT, measured at three layers, shows the opposite trend. In both MaxToki sizes, genes that change more are, weakly, the genes whose next-gene prediction gains more from the cell's own context. How the change is organised differs between architectures. The organisation follows co-expression: in four models, context does not move genes significantly more along functional axes than along axes built from gene sets with the same co-expression. Context moves genes along axes built from co-expressed gene sets much more in the models than in three expression-only representations of the same cells. Among the things we measured, this is what separates the models from expression alone. Two direct tests for context-specific gene function are negative in every model. So we could not read a gene's changed role in a new cell type from these models, which is the use that would make them a tool for new hypotheses. Our tests could detect only clear effects, so this does not prove the models lack it, and whether models trained on more than expression can show it was not tested.

List of abbreviations

AUC: area under the receiver operating characteristic curve; CI: confidence interval; ESM2: Evolutionary Scale Modeling 2; GO: Gene Ontology; PC: principal component; ROC: receiver operating characteristic; SCFM: single-cell foundation model; TF: transcription factor; UCE:

1071 Universal Cell Embeddings.

1072 **Declarations**

1073 **Ethics approval and consent to participate**

1074 Ethics approval and consent to participate were not required for this study, which is a secondary
1075 analysis of publicly available, de-identified human single-cell RNA-sequencing data (Tabula
1076 Sapiens [45] and Setty et al. [50]) and involved no contact with participants. The original
1077 studies obtained ethics approval and informed consent, as described in those publications. No
1078 new human data or samples were collected.

1079 **Consent for publication**

1080 Not applicable.

1081 **Availability of data and materials**

1082 The datasets analysed during the current study are available in the following public repositories:
1083 the Tabula Sapiens immune, kidney and lung datasets in CZ CELLxGENE Discover [47–49];
1084 the human CD34+ bone-marrow data of Setty et al. [50], as distributed with scVelo, in figshare
1085 [52]; the MaxToki checkpoints in Hugging Face [41]; the scGPT whole-human checkpoint [42];
1086 the STATE SE-600M model in Hugging Face [43]; the Geneformer gene dictionaries in Hugging
1087 Face [61]; the GEARS gene-to-GO annotation file in Harvard Dataverse [56]; TRRUST v2 [58];
1088 and DoRotheA [60]. All analysis code, the result files from which every number, table and
1089 figure is computed, the figure code and the list of the 20,000 immune cells used are available
1090 in the gene-context repository [62] and archived at Zenodo (DOI 10.5281/zenodo.23119961)

[63]. The representation tensors for MaxToki-217M, MaxToki-1B, scGPT, STATE, the three untrained models, the three expression-only representations and the Setty data (about 10 GB) are not deposited because of their size. They can be regenerated from the public inputs with the provided extraction scripts and are available from the corresponding author on reasonable request. No new experimental data were generated.

Software availability:

- Project name: gene-context
- Project home page: <https://github.com/Biodyn-AI/gene-context>
- Archived version: Zenodo, DOI 10.5281/zenodo.23119961 (release release2 of the project repository, the same code as tag `bmc-submission-v3.1`) [63]
- Operating system(s): platform independent (developed and run on macOS, Apple silicon)
- Programming language: Python 3.12
- Other requirements: NumPy, SciPy, scikit-learn, h5py, matplotlib and openpyxl; for the extraction scripts, PyTorch (2.2 or later) and transformers (4.43 or later); for scGPT, a local copy of the scGPT source code and the whole-human checkpoint [42] (only its model file is imported) and tqdm; for STATE, the Arc Institute state package and anndata; versions used: PyTorch 2.12.1, transformers 5.13.0, NumPy 2.4
- License: MIT
- Any restrictions to use by non-academics: none

Competing interests

The author declares that he has no competing interests.

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Authors' contributions

IK conceived and designed the study, wrote the software, carried out the analyses, interpreted the results and wrote the manuscript. The author read and approved the final manuscript.

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References

1. Li K, Hopkins AK, Bau D, Viégas F, Pfister H, Wattenberg M. Emergent world representations: Exploring a sequence model trained on a synthetic task. In: The Eleventh International Conference on Learning Representations (ICLR 2023). 2023.
2. Nanda N, Lee A, Wattenberg M. Emergent linear representations in world models of self-supervised sequence models. In: Proceedings of the 6th BlackboxNLP Workshop: Analyzing and Interpreting Neural Networks for NLP. Singapore: Association for Computational Linguistics; 2023. p. 16–30. <https://doi.org/10.18653/v1/2023.blackboxnlp-1.2>.

- 1130 3. Karvonen A. Emergent world models and latent variable estimation in chess-playing language
1131 models. arXiv. 2024. <https://doi.org/10.48550/arXiv.2403.15498>.
- 1132 4. Gurnee W, Tegmark M. Language models represent space and time. In: The Twelfth Inter-
1133 national Conference on Learning Representations (ICLR 2024). 2024.
- 1134 5. Turner AM, Thiergart L, Leech G, Udell D, Vazquez JJ, Mini U, et al. Steering language
1135 models with activation engineering. arXiv. 2023. <https://doi.org/10.48550/arXiv.2308.10248>.
- 1136 6. Zou A, Phan L, Chen S, Campbell J, Guo P, Ren R, et al. Representation engineering: A top-
1137 down approach to AI transparency. arXiv. 2023. <https://doi.org/10.48550/arXiv.2310.01405>.
- 1138 7. Park K, Choe YJ, Veitch V. The linear representation hypothesis and the geometry of large
1139 language models. In: Proceedings of the 41st International Conference on Machine Learning.
1140 Proceedings of Machine Learning Research, vol. 235. PMLR; 2024. p. 39643–66.
- 1141 8. Peters ME, Neumann M, Iyyer M, Gardner M, Clark C, Lee K, et al. Deep contextualized
1142 word representations. In: Proceedings of the 2018 Conference of the North American Chapter
1143 of the Association for Computational Linguistics: Human Language Technologies, Volume 1
1144 (Long Papers). New Orleans: Association for Computational Linguistics; 2018. p. 2227–37.
1145 <https://doi.org/10.18653/v1/N18-1202>.
- 1146 9. Ethayarajh K. How contextual are contextualized word representations? Comparing the ge-
1147 ometry of BERT, ELMo, and GPT-2 embeddings. In: Proceedings of the 2019 Conference on

- 1148 Empirical Methods in Natural Language Processing and the 9th International Joint Conference
1149 on Natural Language Processing (EMNLP-IJCNLP). Hong Kong: Association for Computa-
1150 tional Linguistics; 2019. p. 55–65. <https://doi.org/10.18653/v1/D19-1006>.
- 1151 10. Coenen A, Reif E, Yuan A, Kim B, Pearce A, Viégas F, et al. Visualizing and measuring
1152 the geometry of BERT. In: Advances in Neural Information Processing Systems 32 (NeurIPS
1153 2019). 2019.
- 1154 11. Mu J, Bhat S, Viswanath P. All-but-the-top: Simple and effective postprocessing for word
1155 representations. In: The Sixth International Conference on Learning Representations (ICLR
1156 2018). 2018.
- 1157 12. Cai X, Huang J, Bian Y, Church K. Isotropy in the contextual embedding space: Clusters
1158 and manifolds. International Conference on Learning Representations (ICLR). 2021. <https://openreview.net/forum?id=xYGNO86OWDH>. Accessed 1 Oct 2026.
- 1160 13. Timkey W, van Schijndel M. All bark and no bite: Rogue dimensions in transformer lan-
1161 guage models obscure representational quality. In: Proceedings of the 2021 Conference on Em-
1162 pirical Methods in Natural Language Processing. Punta Cana: Association for Computational
1163 Linguistics; 2021. p. 4527–46. <https://doi.org/10.18653/v1/2021.emnlp-main.372>.
- 1164 14. Theodoris CV, Xiao L, Chopra A, Chaffin MD, Al Sayed ZR, Hill MC, et al. Transfer
1165 learning enables predictions in network biology. Nature. 2023;618:616–24. <https://doi.org/10.1038/s41586-023-06139-9>.
- 1166

- 1167 15. Cui H, Wang C, Maan H, Pang K, Luo F, Duan N, et al. scGPT: Toward building a foundation
1168 model for single-cell multi-omics using generative AI. *Nat Methods*. 2024;21:1470–80. <https://doi.org/10.1038/s41592-024-02201-0>.
1169
- 1170 16. Hao M, Gong J, Zeng X, Liu C, Guo Y, Cheng X, et al. Large-scale foundation model on
1171 single-cell transcriptomics. *Nat Methods*. 2024;21:1481–91. <https://doi.org/10.1038/s41592-024-02305-7>.
1172
- 1173 17. Yang F, Wang W, Wang F, Fang Y, Tang D, Huang J, et al. scBERT as a large-scale pretrained
1174 deep language model for cell type annotation of single-cell RNA-seq data. *Nat Mach Intell*.
1175 2022;4:852–66. <https://doi.org/10.1038/s42256-022-00534-z>.
- 1176 18. Kendiukhov I. Two axes in the gene-embedding space of single-cell foundation models:
1177 What a protein language model already knows, and what requires single-cell pretraining. *SSRN*.
1178 2026. <https://doi.org/10.2139/ssrn.6577900>.
- 1179 19. Bian H, Chen Y, Luo E, Wu X, Hao M, Wei L, et al. General-purpose pre-trained large
1180 cellular models for single-cell transcriptomics. *Natl Sci Rev*. 2024;11:nwae340. <https://doi.org/10.1093/nsr/nwae340>.
1181
- 1182 20. Gilpin W. The cell as a token: High-dimensional geometry in language models and cell
1183 embeddings. *Bioinformatics*. 2025;41:btaf595. <https://doi.org/10.1093/bioinformatics/btaf595>.
1184

- 1185 21. Garí Soler A, Apidianaki M. Let's play mono-poly: BERT can reveal words' polysemy
1186 level and partitionability into senses. *Trans Assoc Comput Linguist.* 2021;9:825–44. https://doi.org/10.1162/tacl_a_00400.
1187
- 1188 22. Templeton A, Conerly T, Marcus J, Lindsey J, Bricken T, Chen B, et al. Scaling monose-
1189 manticity: Extracting interpretable features from Claude 3 Sonnet. *Transformer Circuits Thread*.
1190 2024. <https://transformer-circuits.pub/2024/scaling-monosemanticity/>. Accessed 1 Oct 2026.
- 1191 23. Rosen Y, Roohani Y, Agrawal A, Samotorčan L, Tabula Sapiens Consortium, Quake SR, et al.
1192 Universal cell embedding provides a foundation model for cell biology. *Nature.* 2026;656:183–
1193 91. <https://doi.org/10.1038/s41586-026-10689-z>.
- 1194 24. Adduri AK, Gautam D, Bevilacqua B, Naghipourfar M, Imran A, Shah R, et al. Predicting
1195 cellular responses to perturbation across diverse contexts with State. *Cell.* 2026;189:5914–
1196 5931.e20. <https://doi.org/10.1016/j.cell.2026.07.052>.
- 1197 25. Gómez Ortega J, Nadadur RD, Kunitomi A, Kothen-Hill S, Wagner JUG, Kurtoglu SD, et
1198 al. Temporal AI model predicts drivers of cell state trajectories across human aging. *bioRxiv*.
1199 2026. <https://doi.org/10.64898/2026.03.30.715396>.
- 1200 26. Kendiukhov I. Multi-dimensional spectral geometry of biological knowledge in single-cell
1201 transformer representations. *arXiv.* 2026. <https://doi.org/10.48550/arXiv.2602.22247>.
- 1202 27. Kendiukhov I. What topological and geometric structure do biological foundation models

- 1203 learn? Evidence from 141 hypotheses. PLoS One. 2026;21:e0344826. <https://doi.org/10.1371/>
1204 [journal.pone.0344826](https://doi.org/10.1371/journal.pone.0344826).
- 1205 28. Kendiukhov I. Residual-stream geometry of single-cell foundation models carries incremen-
1206 tal gene-regulatory signal across tissues. BMC Bioinformatics. 2026;27:162. [https://doi.org/10](https://doi.org/10.1186/s12859-026-06538-5)
1207 [.1186/s12859-026-06538-5](https://doi.org/10.1186/s12859-026-06538-5).
- 1208 29. Kedzierska KZ, Crawford L, Amini AP, Lu AX. Zero-shot evaluation reveals limitations of
1209 single-cell foundation models. Genome Biol. 2025;26:101. [https://doi.org/10.1186/s13059-](https://doi.org/10.1186/s13059-025-03574-x)
1210 [025-03574-x](https://doi.org/10.1186/s13059-025-03574-x).
- 1211 30. Boiarsky R, Singh NM, Buendia A, Amini AP, Getz G, Sontag D. Deeper evaluation of a
1212 single-cell foundation model. Nat Mach Intell. 2024;6:1443–6. [https://doi.org/10.1038/s42256-](https://doi.org/10.1038/s42256-024-00949-w)
1213 [024-00949-w](https://doi.org/10.1038/s42256-024-00949-w).
- 1214 31. Ahlmann-Eltze C, Huber W, Anders S. Deep-learning-based gene perturbation effect pre-
1215 diction does not yet outperform simple linear baselines. Nat Methods. 2025;22:1657–61. [https:](https://doi.org/10.1038/s41592-025-02772-6)
1216 [//doi.org/10.1038/s41592-025-02772-6](https://doi.org/10.1038/s41592-025-02772-6).
- 1217 32. Bendidi I, Whitfield S, Kenyon-Dean K, Ben Yedder H, El Mesbahi Y, Noutahi E, et al.
1218 Benchmarking transcriptomics foundation models for perturbation analysis: One PCA still rules
1219 them all. arXiv. 2024. <https://doi.org/10.48550/arXiv.2410.13956>.
- 1220 33. Liu T, Li K, Wang Y, Li H, Zhao H. Evaluating the utilities of foundation models in single-

- 1221 cell data analysis. *Adv Sci.* 2026;13:e14490. <https://doi.org/10.1002/advs.202514490>.
- 1222 34. Kendiukhov I. Sparse autoencoders reveal organized biological knowledge but minimal reg-
1223 ulatory logic in single-cell foundation models: A comparative atlas of Geneformer and scGPT.
1224 arXiv. 2026. <https://doi.org/10.48550/arXiv.2603.02952>.
- 1225 35. Kendiukhov I. Systematic evaluation of single-cell foundation model interpretabil-
1226 ity: Attention-derived edge scores add no incremental value over gene-level features for
1227 perturbation-target prediction. *BMC Genomics.* 2026;27:634. [https://doi.org/10.1186/s12864-](https://doi.org/10.1186/s12864-026-12965-8)
1228 [026-12965-8](https://doi.org/10.1186/s12864-026-12965-8).
- 1229 36. van Dam S, Vösa U, van der Graaf A, Franke L, de Magalhães JP. Gene co-expression analy-
1230 sis for functional classification and gene-disease predictions. *Brief Bioinform.* 2018;19:575–92.
1231 <https://doi.org/10.1093/bib/bbw139>.
- 1232 37. Du J, Jia P, Dai Y, Tao C, Zhao Z, Zhi D. Gene2vec: Distributed representation of genes
1233 based on co-expression. *BMC Genomics.* 2019;20 Suppl 1:82. [https://doi.org/10.1186/s12864-](https://doi.org/10.1186/s12864-018-5370-x)
1234 [018-5370-x](https://doi.org/10.1186/s12864-018-5370-x).
- 1235 38. Chen Y, Zou J. Simple and effective embedding model for single-cell biology built from
1236 ChatGPT. *Nat Biomed Eng.* 2025;9:483–93. <https://doi.org/10.1038/s41551-024-01284-6>.
- 1237 39. Jeffery CJ. Moonlighting proteins. *Trends Biochem Sci.* 1999;24:8–11. [https://doi.org/10](https://doi.org/10.1016/S0968-0004(98)01335-8)
1238 [.1016/S0968-0004\(98\)01335-8](https://doi.org/10.1016/S0968-0004(98)01335-8).

- 1239 40. Heinz S, Benner C, Spann N, Bertolino E, Lin YC, Laslo P, et al. Simple com-
1240 binations of lineage-determining transcription factors prime cis-regulatory elements
1241 required for macrophage and B cell identities. *Mol Cell*. 2010;38:576–89. <https://doi.org/10.1016/j.molcel.2010.05.004>.
1242
- 1243 41. Theodoris Lab. MaxToki model checkpoints (217M and 1B, Hugging Face safetensors).
1244 Hugging Face. 2026. <https://huggingface.co/theodoris-lab/MaxToki>. Accessed 1 Oct 2026.
- 1245 42. Bo Wang Lab. scGPT source code and pretrained whole-human checkpoint. GitHub. <https://github.com/bowang-lab/scGPT>. Accessed 1 Oct 2026.
1246
- 1247 43. Arc Institute. State Embedding model SE-600M. Hugging Face. 2025. <https://huggingface.co/arcinstitute/SE-600M>. Accessed 1 Oct 2026.
1248
- 1249 44. Lin Z, Akin H, Rao R, Hie B, Zhu Z, Lu W, et al. Evolutionary-scale prediction of atomic-
1250 level protein structure with a language model. *Science*. 2023;379:1123–30. [https://doi.org/10](https://doi.org/10.1126/science.adc2574)
1251 [.1126/science.adc2574](https://doi.org/10.1126/science.adc2574).
- 1252 45. Tabula Sapiens Consortium. The Tabula Sapiens: A multiple-organ, single-cell transcrip-
1253 tomic atlas of humans. *Science*. 2022;376:eabl4896. <https://doi.org/10.1126/science.abl4896>.
- 1254 46. CZI Cell Science Program, Abdulla S, Aeversmann B, Assis P, Badajoz S, Bell SM, et al.
1255 CZ CELLxGENE Discover: A single-cell data platform for scalable exploration, analysis and
1256 modeling of aggregated data. *Nucleic Acids Res*. 2025;53:D886–900. <https://doi.org/10.1093/>

1257 nar/gkael142.

1258 47. Tabula Sapiens Consortium. Tabula Sapiens – immune (dataset version e62b0182-368f-
1259 4875-8ac1-f49e3d5eda60, schema 7.0.0). CZ CELLxGENE Discover. [https://datasets.cellxgene](https://datasets.cellxgene.ne.cziscience.com/e62b0182-368f-4875-8ac1-f49e3d5eda60.h5ad)
1260 [ne.cziscience.com/e62b0182-368f-4875-8ac1-f49e3d5eda60.h5ad](https://datasets.cellxgene.ne.cziscience.com/e62b0182-368f-4875-8ac1-f49e3d5eda60.h5ad). Accessed 1 Oct 2026.

1261 48. Tabula Sapiens Consortium. Tabula Sapiens – kidney (dataset version 65ca6e36-73b0-4c88-
1262 b0f3-7b23b48844ad, schema 7.0.0). CZ CELLxGENE Discover. [https://datasets.cellxgene.czi](https://datasets.cellxgene.czi-science.com/65ca6e36-73b0-4c88-b0f3-7b23b48844ad.h5ad)
1263 [science.com/65ca6e36-73b0-4c88-b0f3-7b23b48844ad.h5ad](https://datasets.cellxgene.czi-science.com/65ca6e36-73b0-4c88-b0f3-7b23b48844ad.h5ad). Accessed 1 Oct 2026.

1264 49. Tabula Sapiens Consortium. Tabula Sapiens – lung (dataset version 40f8b1a3-9f76-4ac4-
1265 8761-32078555ed4e, schema 7.0.0). CZ CELLxGENE Discover. [https://datasets.cellxgene.czi](https://datasets.cellxgene.czi-science.com/40f8b1a3-9f76-4ac4-8761-32078555ed4e.h5ad)
1266 [science.com/40f8b1a3-9f76-4ac4-8761-32078555ed4e.h5ad](https://datasets.cellxgene.czi-science.com/40f8b1a3-9f76-4ac4-8761-32078555ed4e.h5ad). Accessed 1 Oct 2026.

1267 50. Setty M, Kisieliovas V, Levine J, Gayoso A, Mazutis L, Pe’er D. Characterization of cell
1268 fate probabilities in single-cell data with Palantir. Nat Biotechnol. 2019;37:451–60. [https:](https://doi.org/10.1038/s41587-019-0068-4)
1269 [//doi.org/10.1038/s41587-019-0068-4](https://doi.org/10.1038/s41587-019-0068-4).

1270 51. Bergen V, Lange M, Peidli S, Wolf FA, Theis FJ. Generalizing RNA velocity to transient
1271 cell states through dynamical modeling. Nat Biotechnol. 2020;38:1408–14. [https://doi.org/10](https://doi.org/10.1038/s41587-020-0591-3)
1272 [.1038/s41587-020-0591-3](https://doi.org/10.1038/s41587-020-0591-3).

1273 52. scVelo project. Human CD34+ bone marrow data of Setty et al. As distributed with scVelo
1274 (bonemarrow.h5ad; figshare file 27686835). figshare. <https://ndownloader.figshare.com/files/2>

1275 7686835. Accessed 1 Oct 2026.

1276 53. Ashburner M, Ball CA, Blake JA, Botstein D, Butler H, Cherry JM, et al. Gene Ontology:
1277 Tool for the unification of biology. *Nat Genet.* 2000;25:25–9. <https://doi.org/10.1038/75556>.

1278 54. Gene Ontology Consortium, Aleksander SA, Balhoff J, Carbon S, Cherry JM, Drabkin
1279 HJ, et al. The Gene Ontology knowledgebase in 2023. *Genetics.* 2023;224:iyad031. [https:](https://doi.org/10.1093/genetics/iyad031)
1280 [//doi.org/10.1093/genetics/iyad031](https://doi.org/10.1093/genetics/iyad031).

1281 55. Roohani Y, Huang K, Leskovec J. Predicting transcriptional outcomes of novel multigene
1282 perturbations with GEARS. *Nat Biotechnol.* 2024;42:927–35. [https://doi.org/10.1038/s41587-](https://doi.org/10.1038/s41587-023-01905-6)
1283 [023-01905-6](https://doi.org/10.1038/s41587-023-01905-6).

1284 56. Roohani Y, Huang K, Leskovec J. GEARS gene-to-Gene Ontology annotation
1285 file (gene2go.pkl; harvard dataverse file 6153417). Harvard Dataverse. 2022. [https:](https://dataverse.harvard.edu/api/access/datafile/6153417)
1286 [//dataverse.harvard.edu/api/access/datafile/6153417](https://dataverse.harvard.edu/api/access/datafile/6153417). Accessed 1 Oct 2026.

1287 57. Han H, Cho J-W, Lee S, Yun A, Kim H, Bae D, et al. TRRUST v2: An expanded refer-
1288 ence database of human and mouse transcriptional regulatory interactions. *Nucleic Acids Res.*
1289 2018;46:D380–6. <https://doi.org/10.1093/nar/gkx1013>.

1290 58. Han H, Cho J-W, Lee S, Yun A, Kim H, Bae D, et al. TRRUST v2 human transcriptional
1291 regulatory interactions (trrust_rawdata.human.tsv). GRNpedia. 2018. [https://www.grnpedia.o](https://www.grnpedia.org/trrust/)
1292 [rg/trrust/](https://www.grnpedia.org/trrust/). Accessed 1 Oct 2026.

- 1293 59. Garcia-Alonso L, Holland CH, Ibrahim MM, Turei D, Saez-Rodriguez J. Benchmark and
1294 integration of resources for the estimation of human transcription factor activities. *Genome Res.*
1295 2019;29:1363–75. <https://doi.org/10.1101/gr.240663.118>.
- 1296 60. Saez-Rodriguez Lab. DoRothEA human regulons. GitHub. <https://github.com/saezlab/dor>
1297 [othea](https://github.com/saezlab/dor). Accessed 1 Oct 2026.
- 1298 61. Theodoris CV. Geneformer: Gene name, gene median and token dictionaries. Hugging Face.
1299 <https://huggingface.co/ctheodoris/Geneformer>. Accessed 1 Oct 2026.
- 1300 62. Kendiukhov I. gene-context: Analysis code, result files and figures for this study. GitHub.
1301 2026. <https://github.com/Biodyn-AI/gene-context>. Accessed 1 Oct 2026.
- 1302 63. Kendiukhov I. gene-context: Analysis code, result files and figures for this study (release2).
1303 2026. <https://doi.org/10.5281/zenodo.23119961>.

1304 **Figures**

1305 **Figure 1. Gene-specific context response from the embedding through all 11 layers of**
1306 **MaxToki-217M.** Agreement of the same gene’s relative context shift across independent halves
1307 of the cells (blue; full depth scan with a 4000-gene panel) against the gene-shuffled null (grey).
1308 Open diamonds mark the four layers of the main 6000-gene extraction (Table 1). Layer 0, the
1309 context-free embedding output, is exactly zero; the response is highest at layer 1 (+0.847) and
1310 falls to +0.651 at layer 11. Values are pooled over 66 cell-type pairs per layer; gene-level boot-
1311 strap 95% CIs are narrower than the markers.

Figure 2. The gene-specific context response is not explained by rank position. EXCESS

(the agreement, as a cosine, of a gene's relative context shift between two independent halves of the cells, minus the same agreement for a randomly paired gene; Table 1) in MaxToki-217M (main extraction, 1000 cells per cell type) at layers 4, 8 and 11 for genes whose rank changes little between cell types (rank-stable, below the median rank change), genes whose rank changes a lot (rank-moving, above the median), and all genes after a linear regression of the shift on the gene's ranks and rank change. The response is about the same for rank-stable and rank-moving genes, and 99%, 98% and 97% of it survives the rank regression at layers 4, 8 and 11 (a regression with curved rank terms gives almost the same; Results).

Figure 3. Functional organisation compared with random and co-expression nulls. For

each of the three functional axes with well-separated poles (MaxToki-217M, main extraction with 1000 cells per cell type, layer 4; the cell-cycle/differentiation axis is omitted), the red marker and line show the functional axis's rank-controlled reproducible interaction power. Boxes show 300 null axes each, all with the functional poles' sizes: random genes (grey); gene sets matched on size and average co-expression (light blue); and modules of the genes most co-expressed with a random seed gene (dark blue), which are 3–11 times more co-expressed than the functional poles. Boxes show the interquartile range and median; whiskers reach the most extreme null value within 1.5 times the interquartile range (values beyond are not drawn). Annotated p values are empirical one-sided values against the matched null and against the strongly co-expressed modules. The functional axes are far above random axes but within both co-expression nulls.

Figure 4. The nuclear/surface direction is read by the model. We added the nuclear/surface

direction at the layer-4 representation of half of each cell's genes (MaxToki-217M, 30 immune cells) and read the model's logits at the other, untouched half. The y axis is the signed swing: the specificity (the change in mean logit of nuclear-pole genes minus that of surface-pole genes)

after a push toward the nuclear pole, minus the specificity after a push toward the surface pole. The x axis is steering strength, in multiples of the mean residual-stream norm. Blue: functional direction (mean $\pm$ standard error over 30 cells). Grey: random direction of the same norm, drawn fresh for each cell. Only the nuclear/surface direction is shown. In the same test, the mitochondrion/cytoskeleton direction has no significant effect at strengths 0.25 and 0.5, and the transcription/transport direction has a reversed effect (Results).

Figure 5. Contextualisation strength and functional organisation of every representation on the same cells. Left: EXCESS (agreement of the same gene's context shift between two halves of the cells, minus that for a randomly paired gene; as in Fig. 2); values of different architectures should not be ranked against each other, because widths and input types differ. Right: functional-z for the nuclear/surface axis against 1000 random-gene axes (no rank regression; genes count-balanced in at least 6 of the 12 cell types). Blue: trained models at depth-matched layers (MaxToki-217M layer 4, MaxToki-1B layer 7, scGPT layer 4, STATE layer 6). Grey: untrained versions of the same architectures (the untrained STATE uses the same fixed protein-embedding gene table as the trained model; only its other weights are random). Orange: expression-only representations built from the same cells without a model. All use the same 600 cells per cell type, partitions and 5000-gene panel; model and centroid vectors average 50 occurrences or expressing cells per entry, and the landmark and principal-component versions all 300 cells of each half. The dashed line marks $z = 3$.

Additional files

Additional file 1 (Excel workbook, .xlsx). Full results of the multi-model comparison. Sheets: EXCESS with the main-effect replication for every representation and layer (trained models at two or three layers, untrained models, expression-only representations), with the rank control

1359 at each trained model's depth-matched layer and for the expression-only representations; func-
1360 tional axes with the random, matched co-expression, strong co-expression and tightness nulls,
1361 for genes count-balanced in at least 6 and at least 9 cell types, with MaxToki rank as the abun-
1362 dance covariate, and the functional axes and co-expression nulls also with mean log-normalised
1363 expression as the covariate (untrained scGPT and STATE: functional axes only); directional con-
1364 gruence and all three curated-target tests, including the untrained models; the comparison of all
1365 representations on each model's own entries and on the shared entries; every steering configura-
1366 tion with both controls (including the 1000-cell MaxToki-217M runs); and every prediction-link
1367 run. It also lists two other builds of the centroid version: with all expressing cells instead of 50
1368 (a check on sample size), and the first build with 512 components, which failed the positive
1369 control and is not used.